

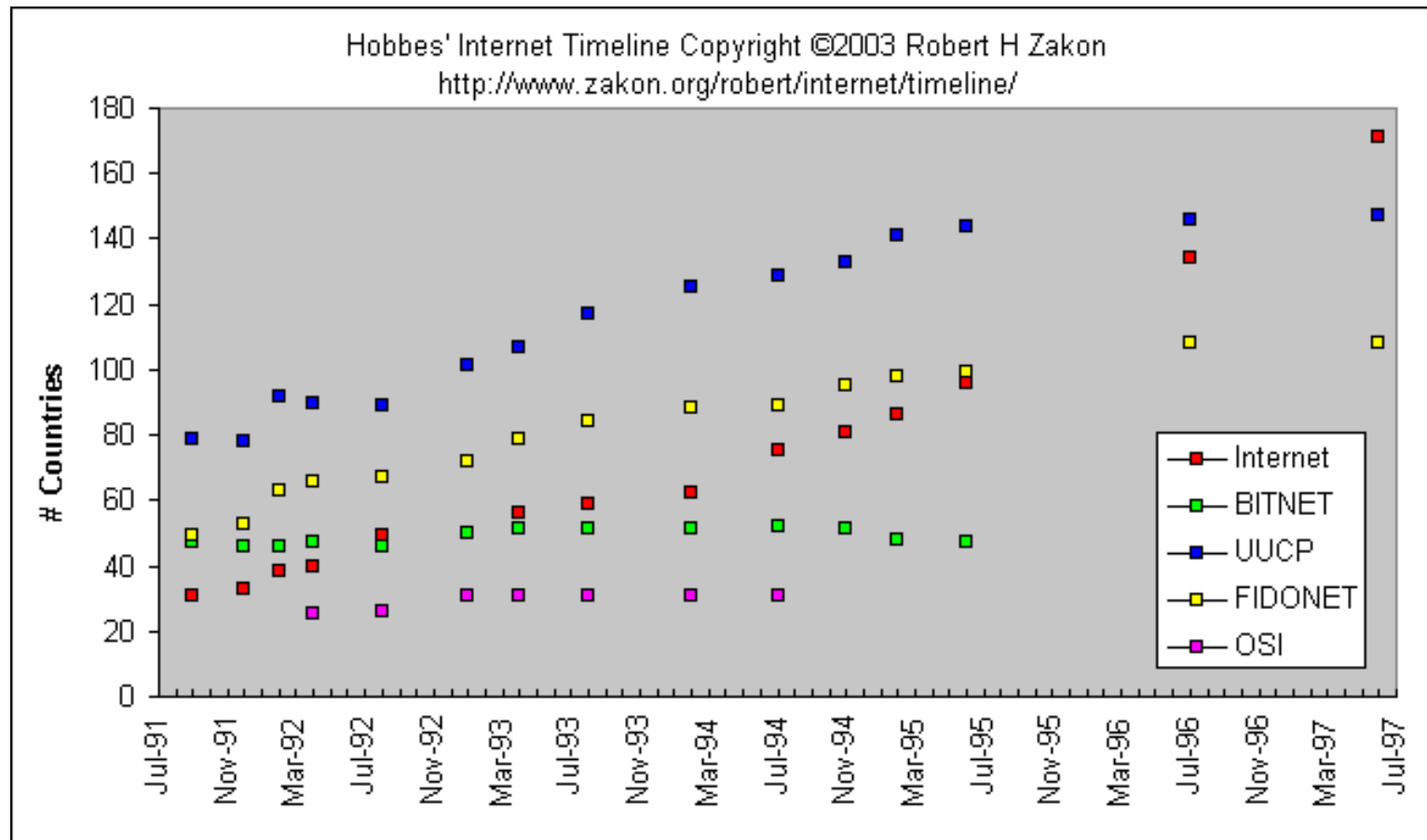
ETS052

Networks IRL

Jens A Andersson



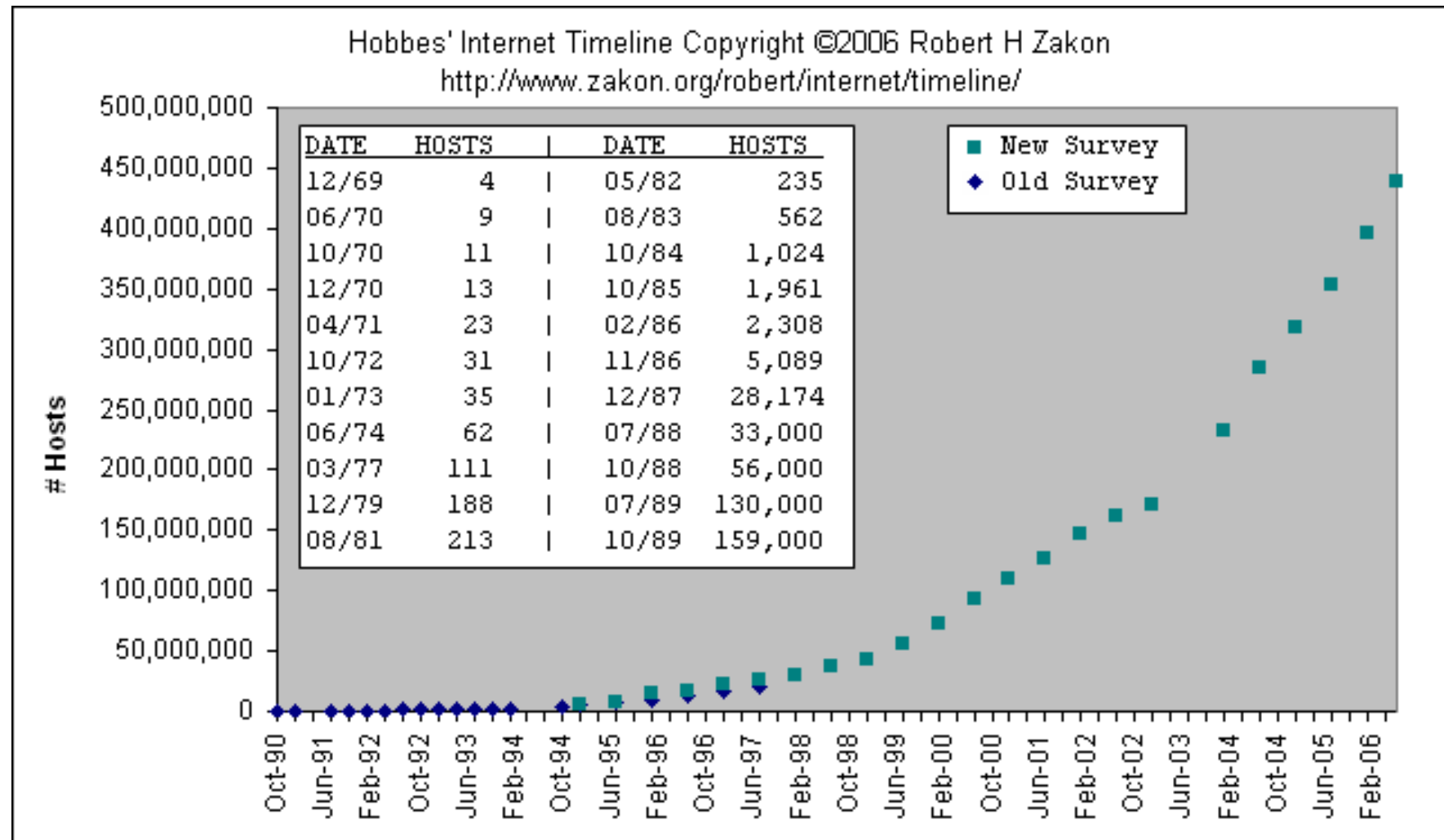
Networks in the World



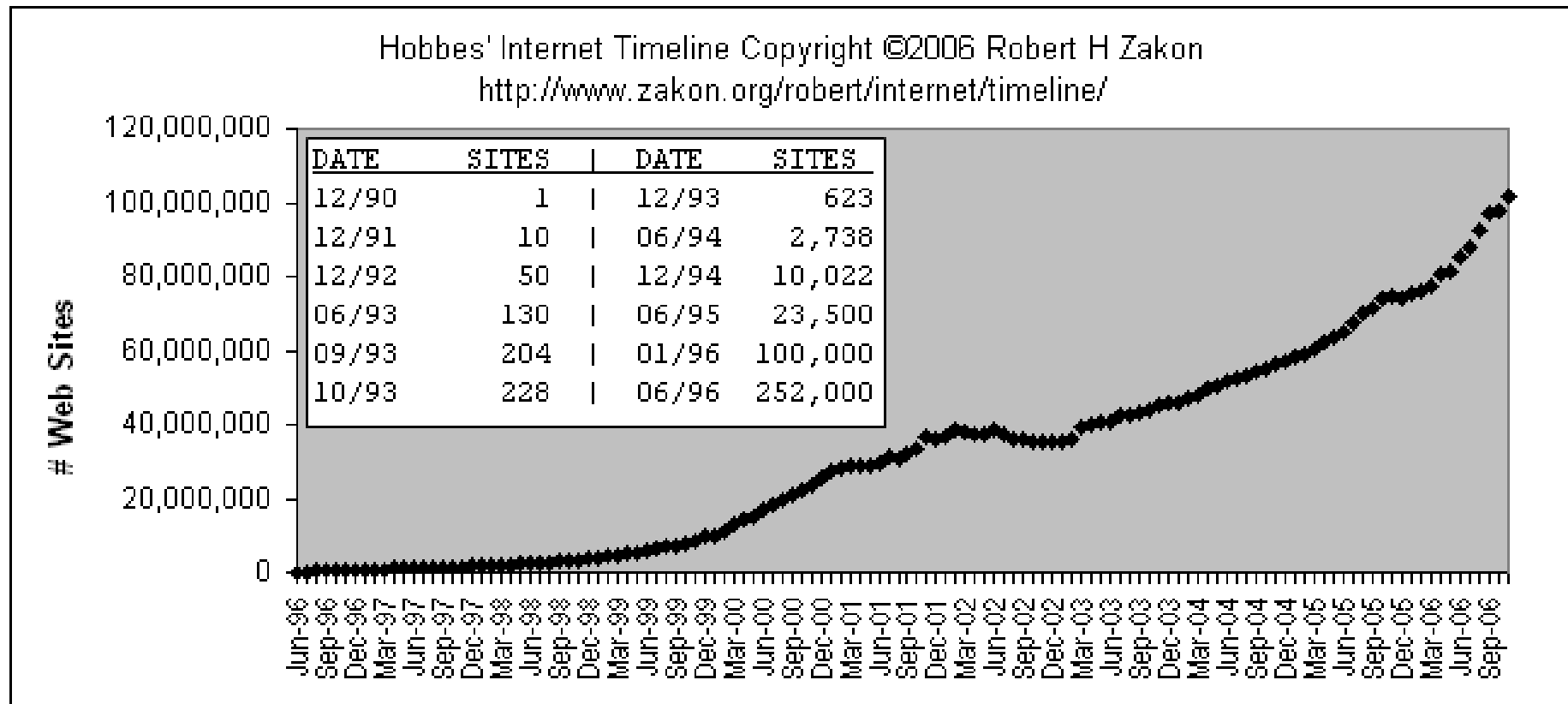
History: Internet

- 1961-69 Research packet switching
- 1969 ARPANET operating
- 1971 Email “invented”
- 1974 Cerf & Kahn publish spec for TCP (which then included “IP”)
- 1978 TCP/IP split up in TCP and IP. UDP added.
- 1983 All of **ARPANET changes to IP/TCP/UDP 1/1 09.00.**
- 1984 ARPANET split. MILNET added. Connected with router. DNS in use.
- 1986 **NFSNET backbone in USA.**
- 1987 First INTEROP.
- 1990 ARPANET closed.
- 1991 Berners-Lee creates WWW.
- 1993 **Web browser Mosaic conkers world**
- 1995 NFSNET back to research network only. Commercial backbones available.
- 1996 Internet2 founded.
- 2002 GRTN Global Terabit Research Network
- 2004 CERNET2 first Chinese IPv6 backbone opens up

Internet: Number of hosts



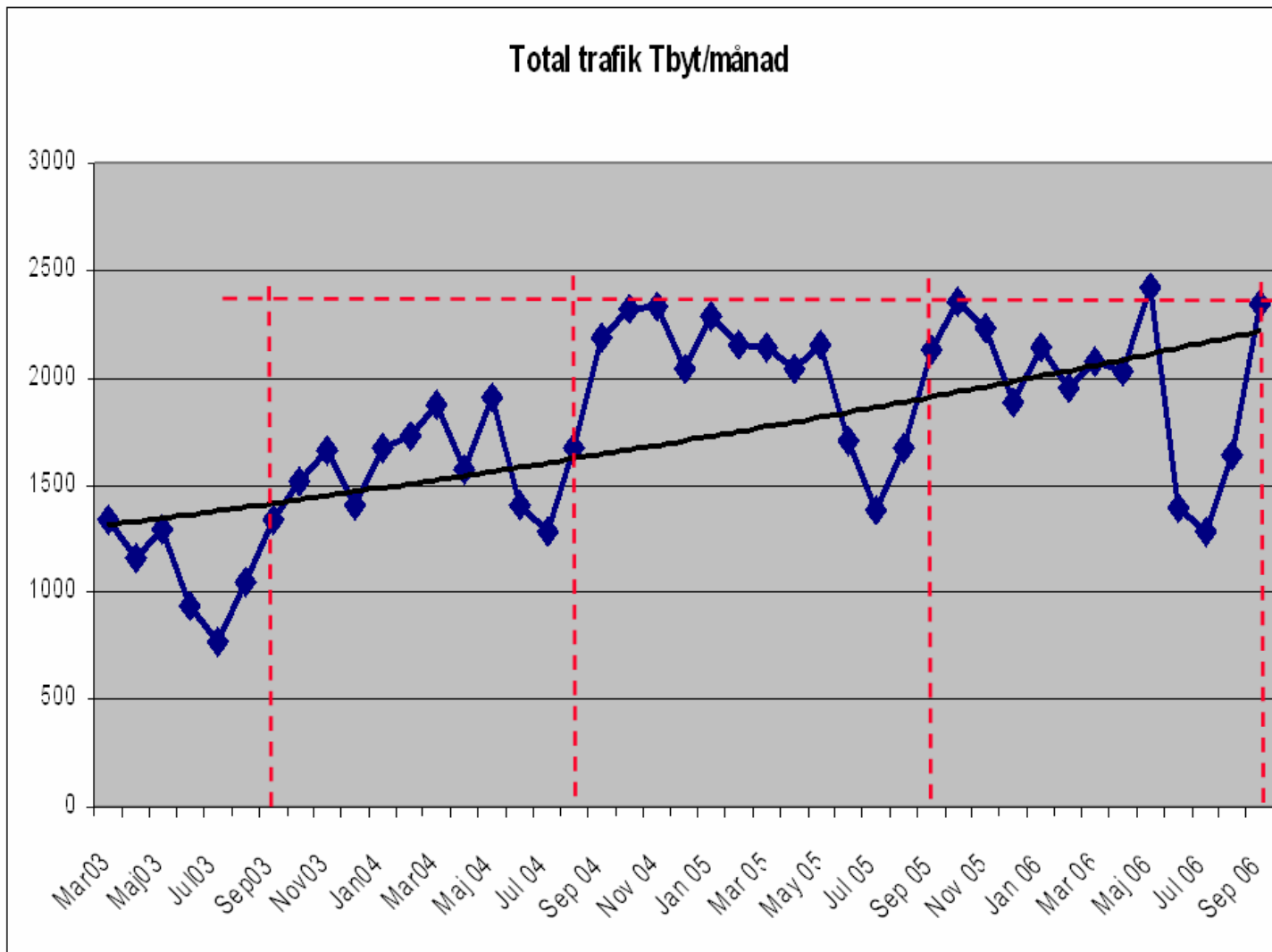
Internet: Number of WWW sites



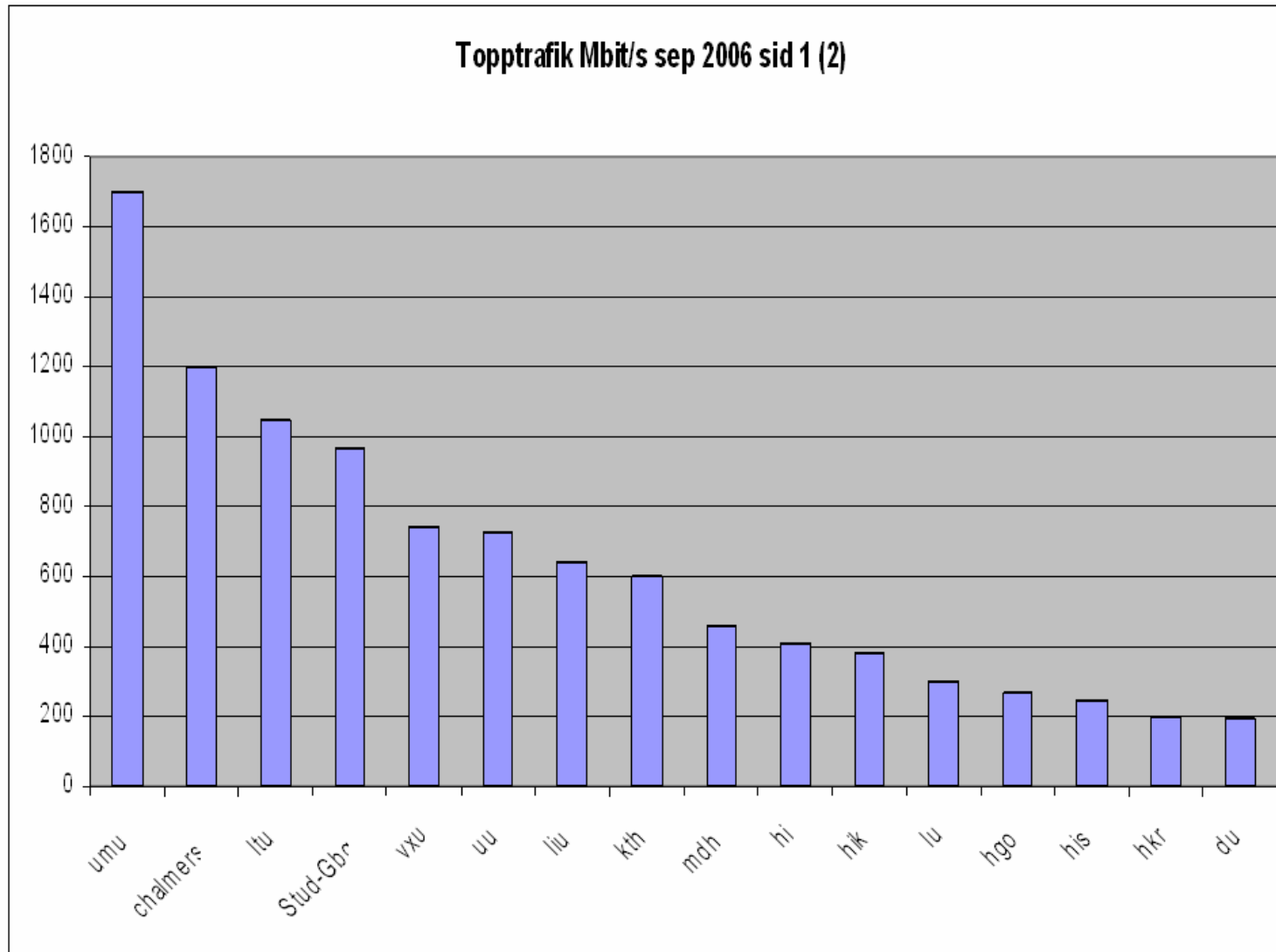
History: LUNET and SUNET

- 1956 SMIL up and running
- 1970 SMIL replaced by Univac
- 1981-82 Terminal network covering Sweden built, called SUNET.
- 1982 Start building terminal network in Lund, LUNET.
- 1982-83 LUNET connected to SUNET. SUNET now network of terminal networks.
- 1984 LDC and E building builds separate Ethernets.
- 1985 Optical fibre connects the two Ethernets. lth.se and lu.se are registered.
- 1986 SUNET becomes a DECnet.
- 1988 Backbone (ethernet + optical fibre) begin install. **SUNET runs IP. Lund connected to Internet.**
- 1991-92 SUNET upgraded.
- 1992-94 Lunds terminal network closed.
- 1994 SUNET upgraded.
- 1997 LUNET upgrade starts.
- 1998 SUNET-155 in operation.
- 2002 Giga-SUNET in operation.
- 2006 GigaLUNET and OptoSUNET in operation

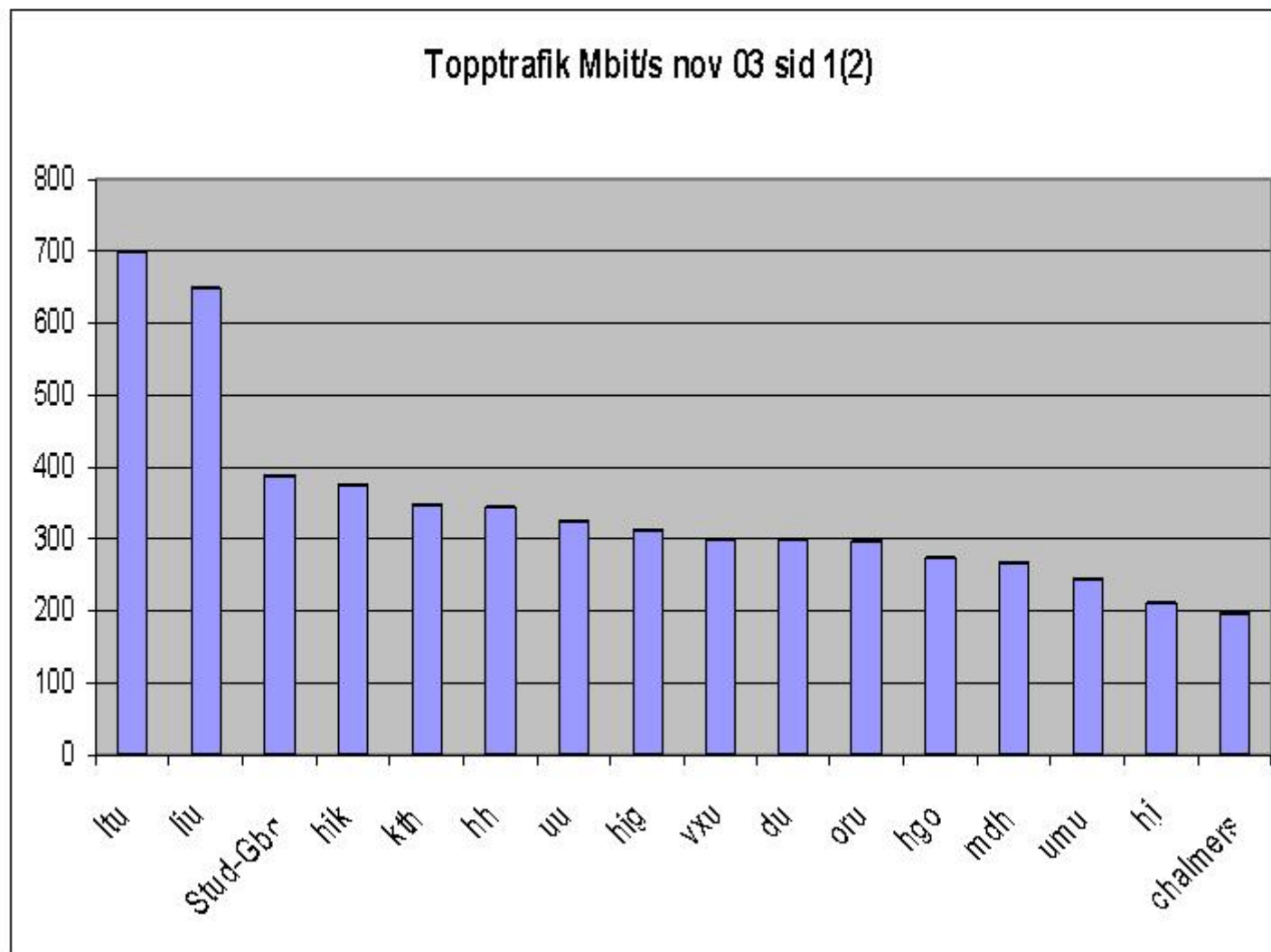
SUNET Total Traffic



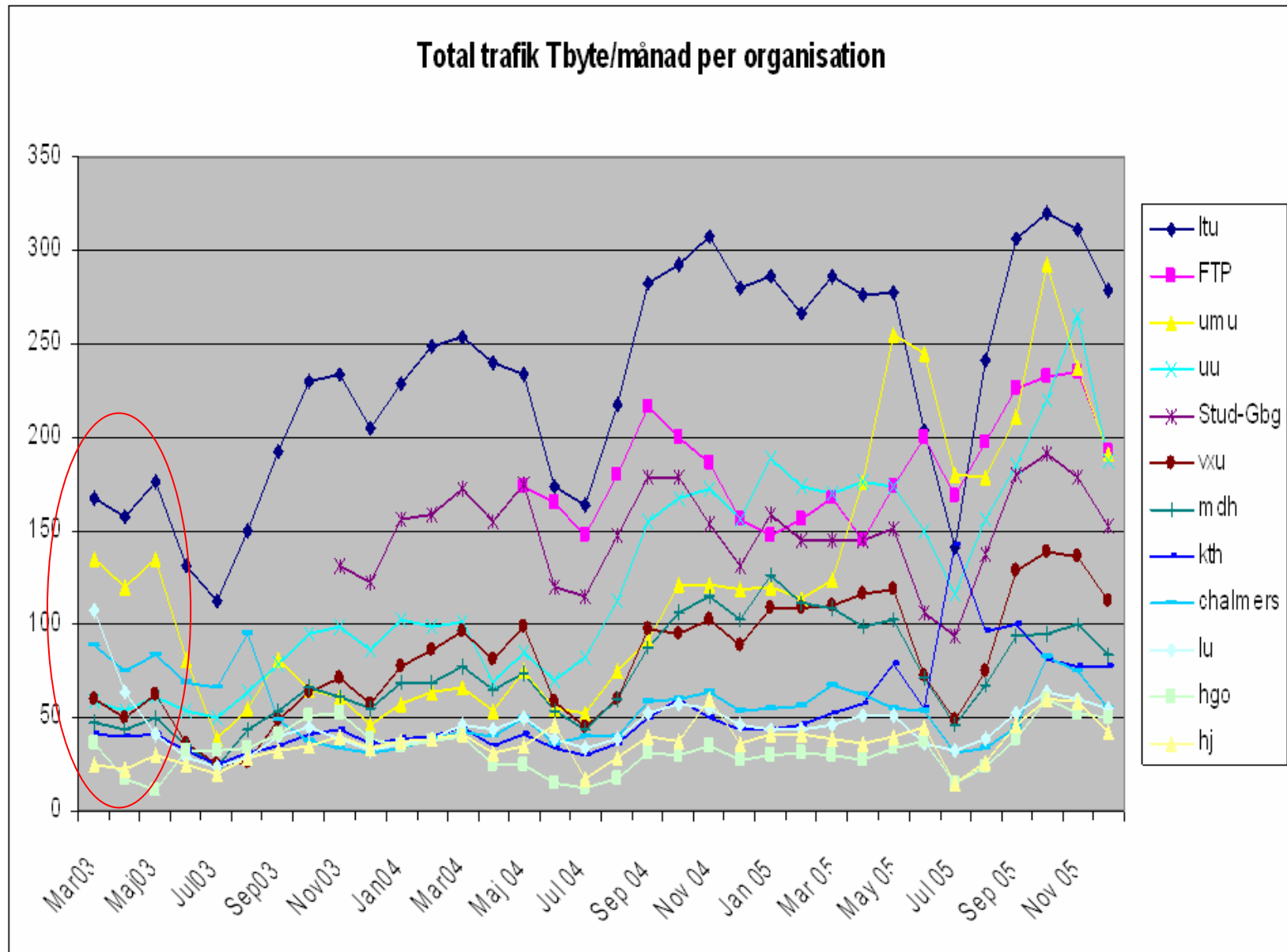
SUNET top traffic sept -06



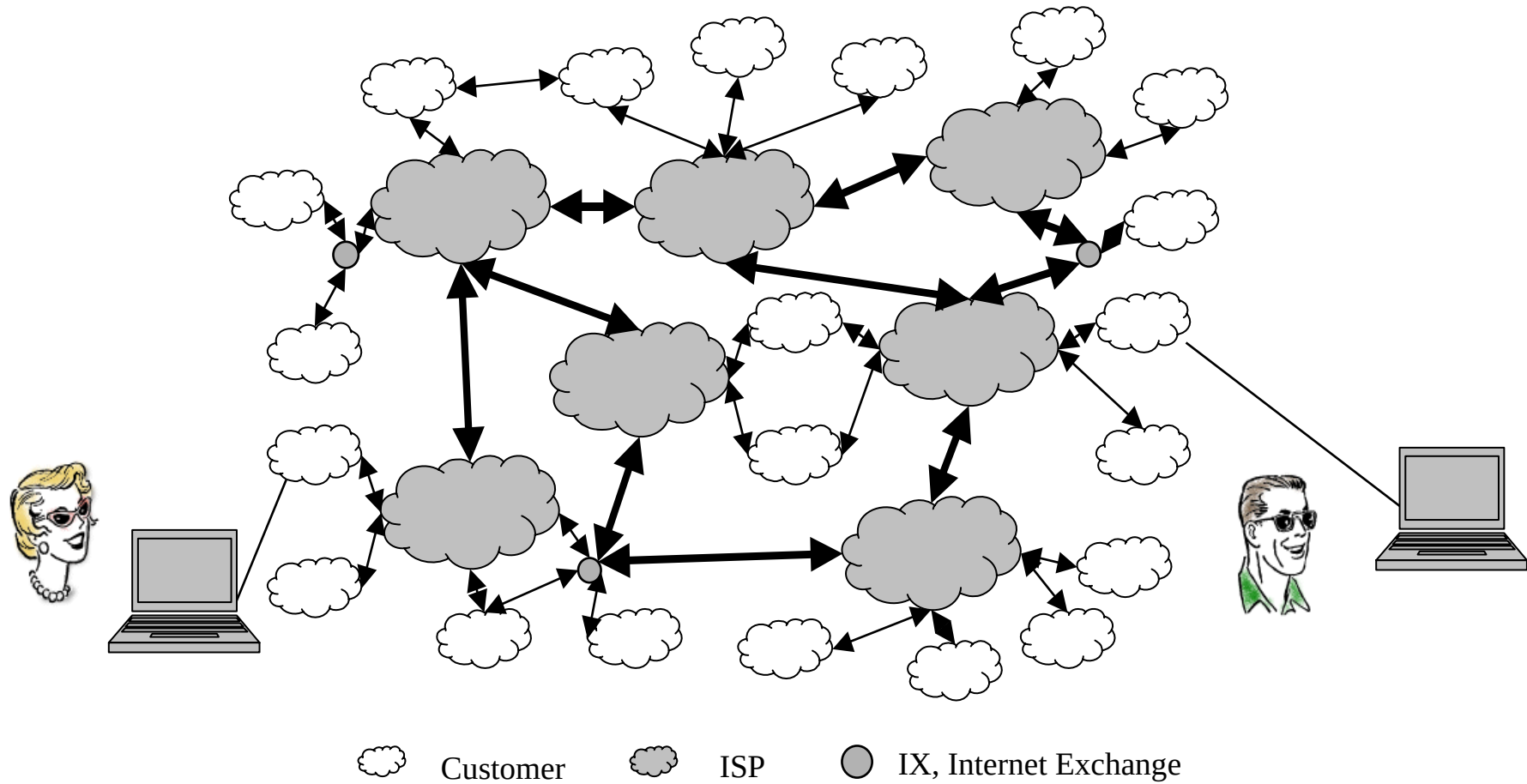
SUNET top traffic nov -03



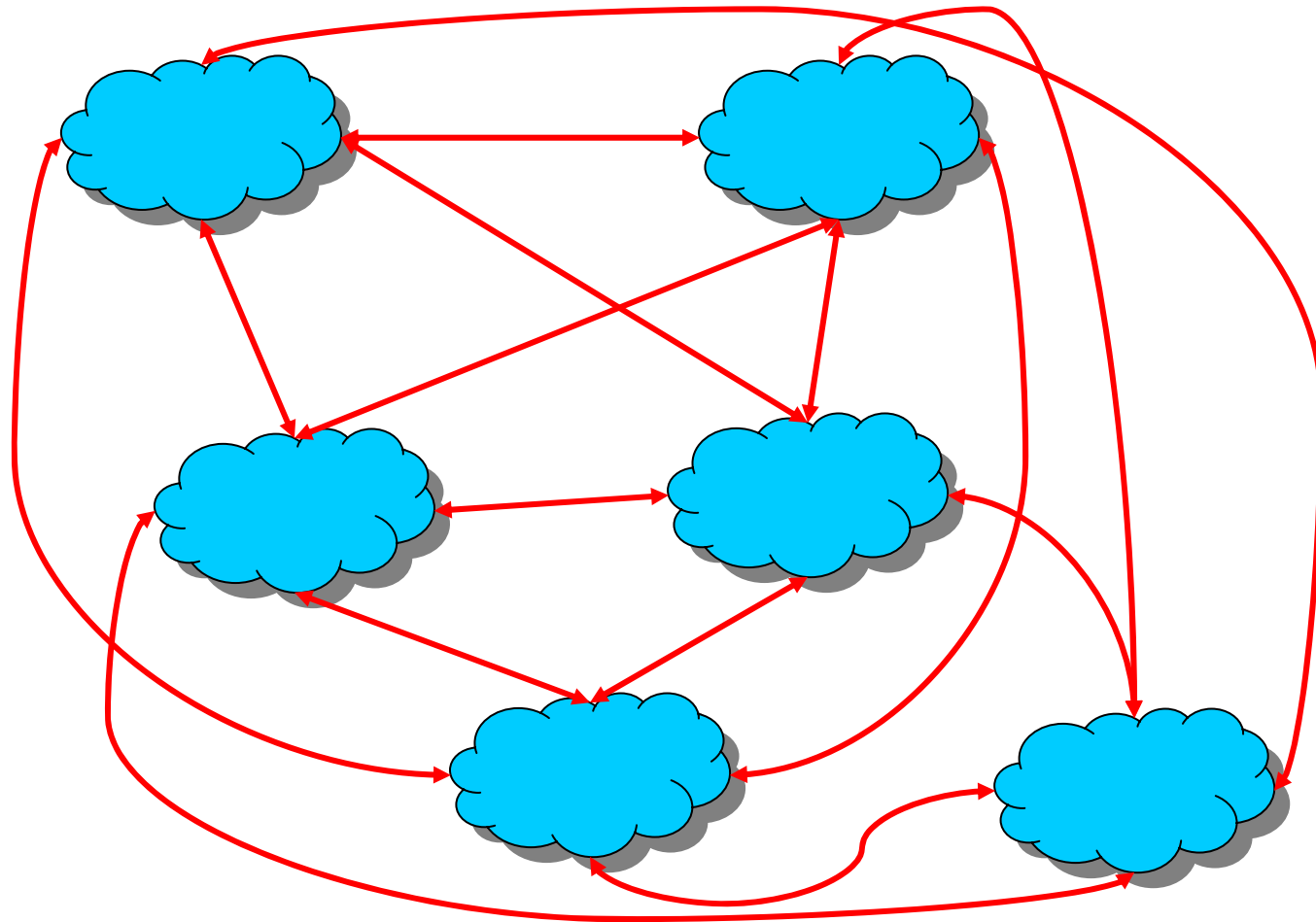
SUNET Total Traffic



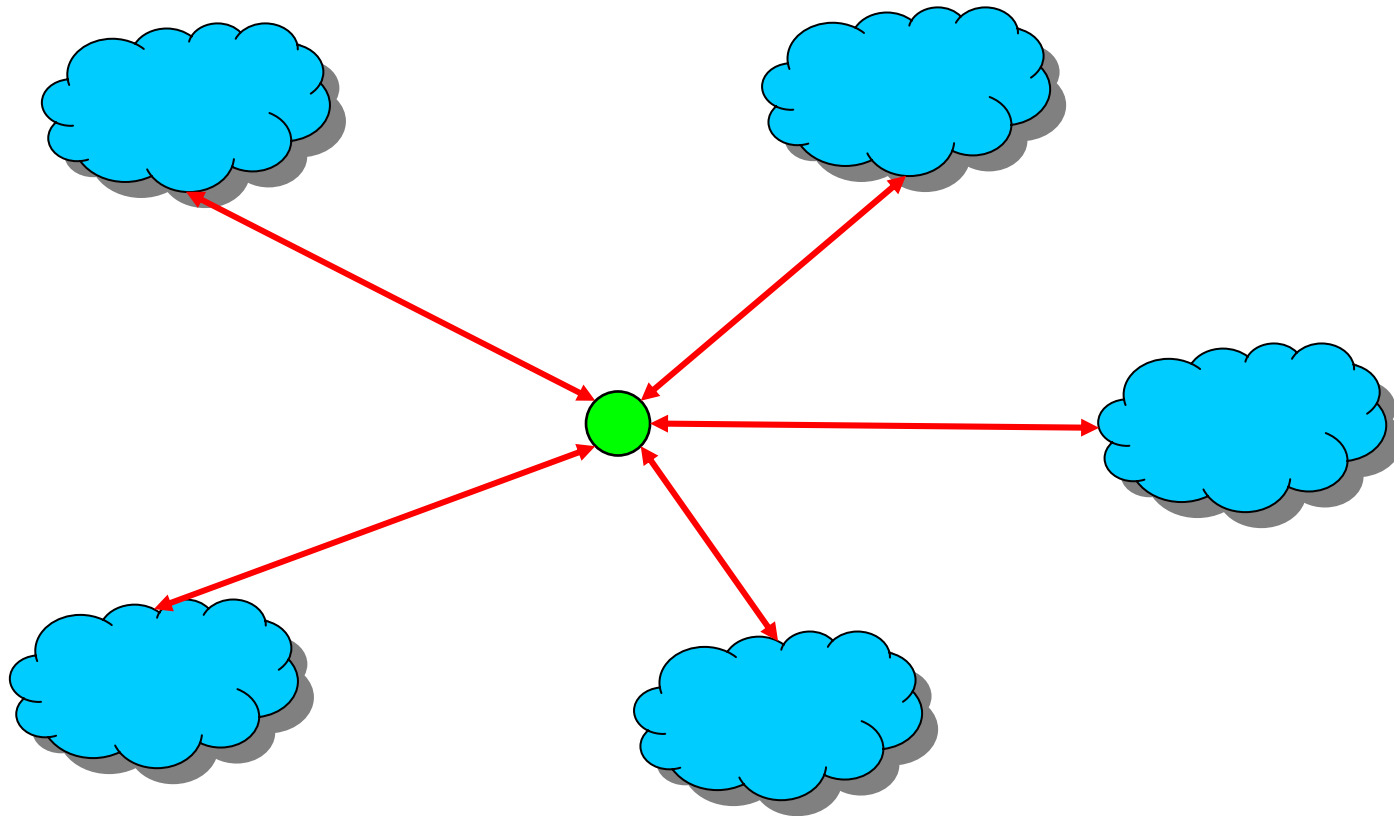
Structure of Internet



Traffic exchange, Directly



Traffic Exchange, in IX

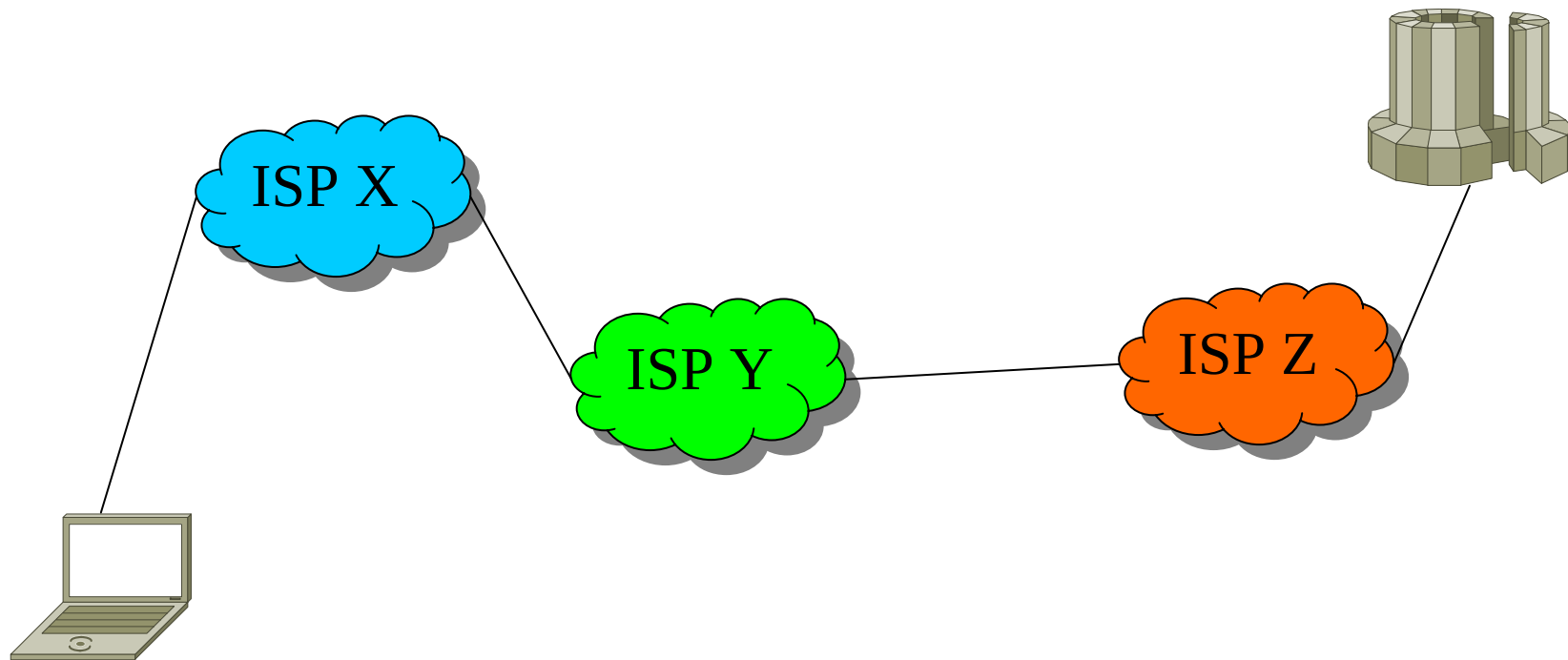


Internet a network of networks where...

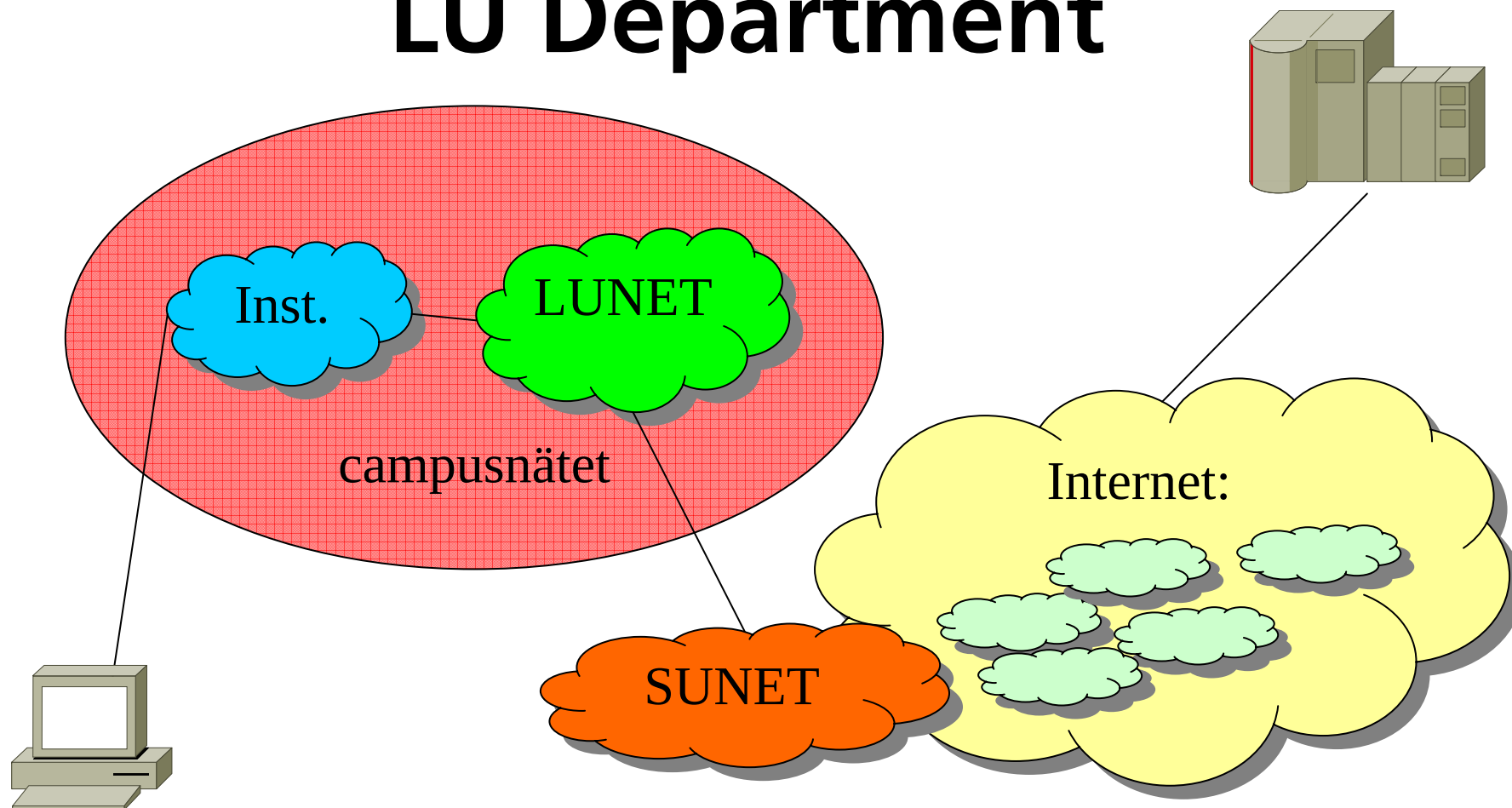
- ... all use same protocol suite, TCP/IP
- ... bilateral agreements controls traffic exchange
- ... traffic exchange performed directly between parties or in global/national/regional/local Internet Exchanges, IX
- QoS?
 - Best Effort

End-to-end performance

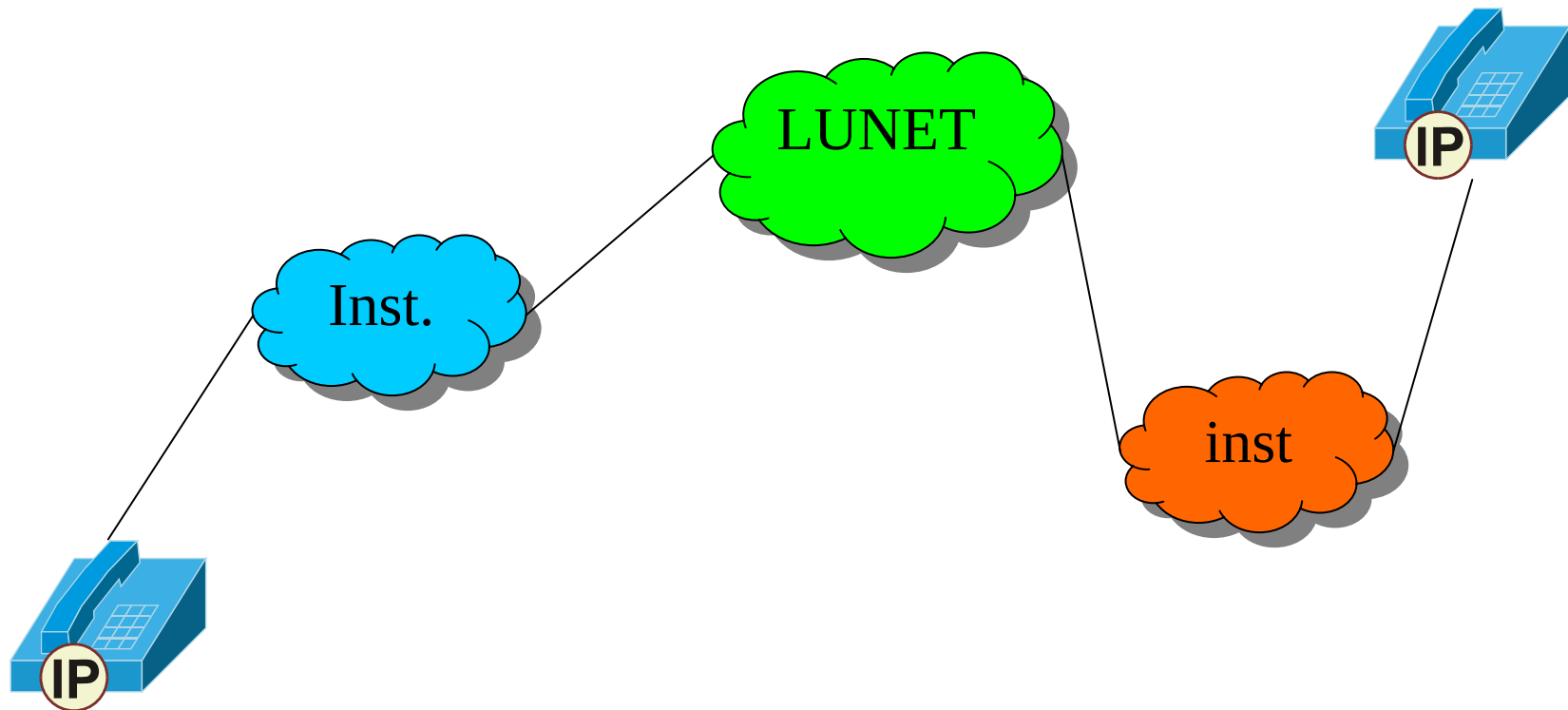
The user perspective!



LU Department



LU IP telephony



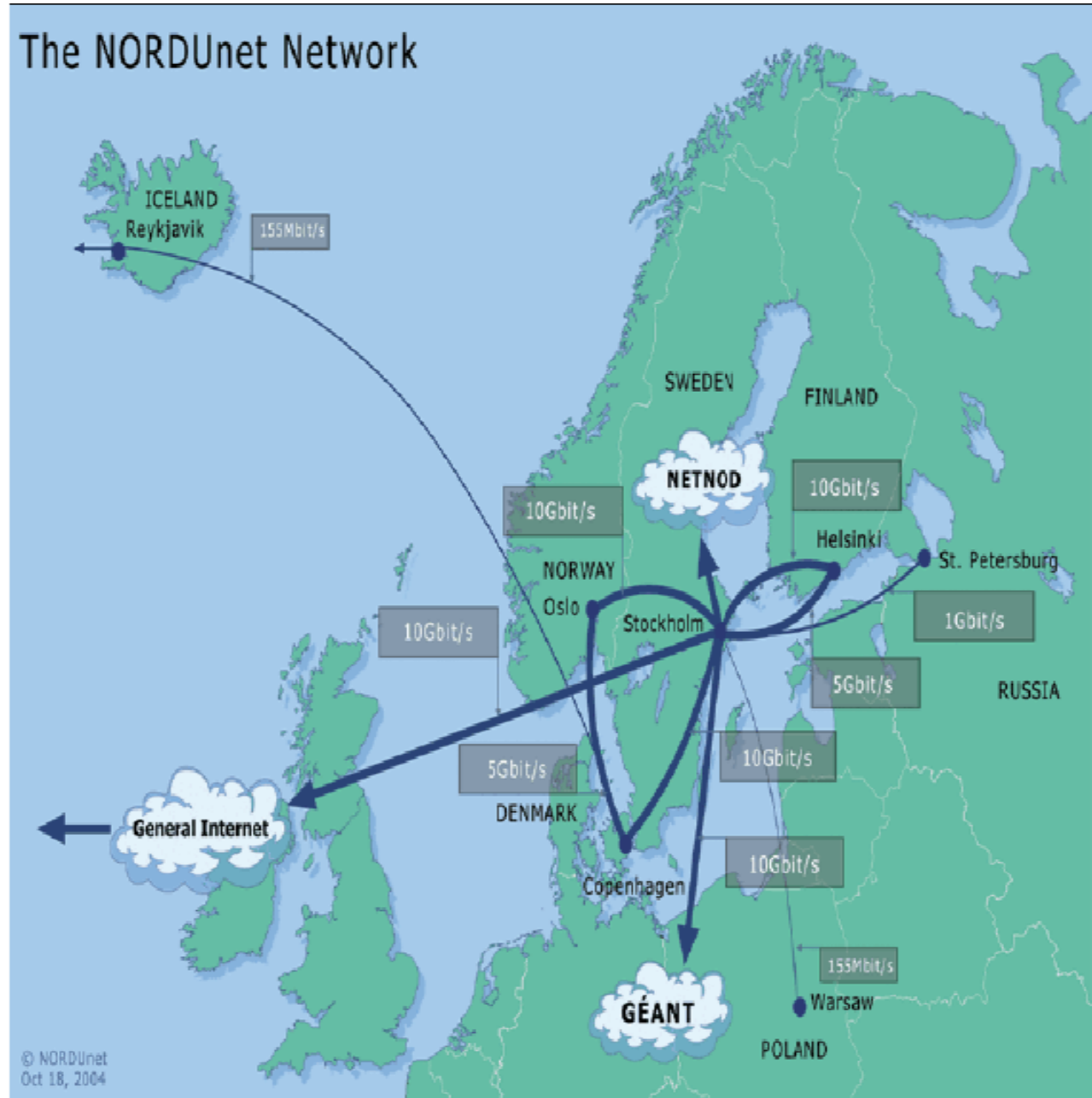
GLIF

(Global Lambda Integrated Facility)



NORDUnet

The NORDUnet Network



GigaSUNET backbone

GigaSunet fjärrnät

NOTERA: Kiruna och Visby är formellt accessnät!

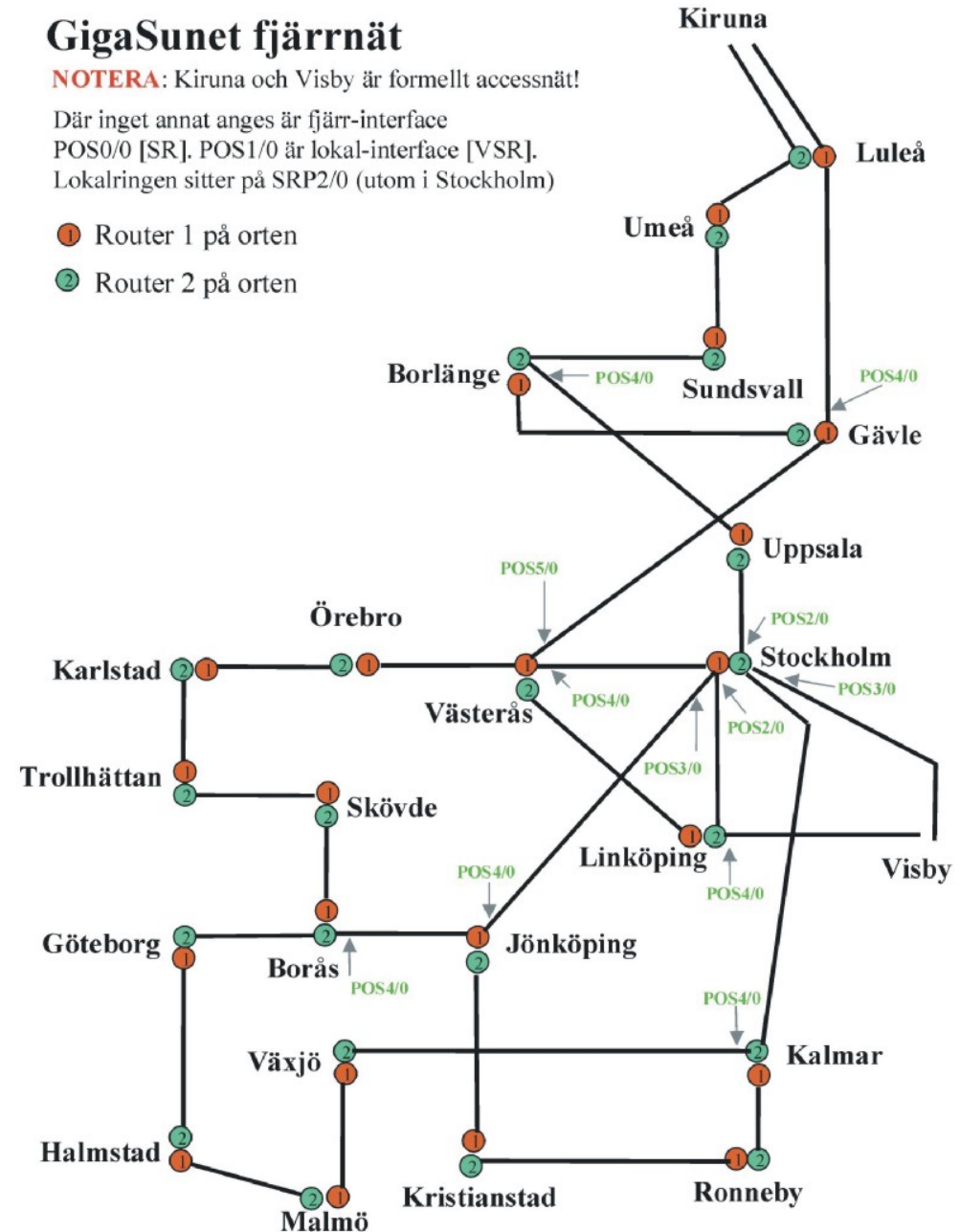
Där inget annat anges är fjärr-interface

POS0/0 [SR], POS1/0 är lokal-interface [VSR].

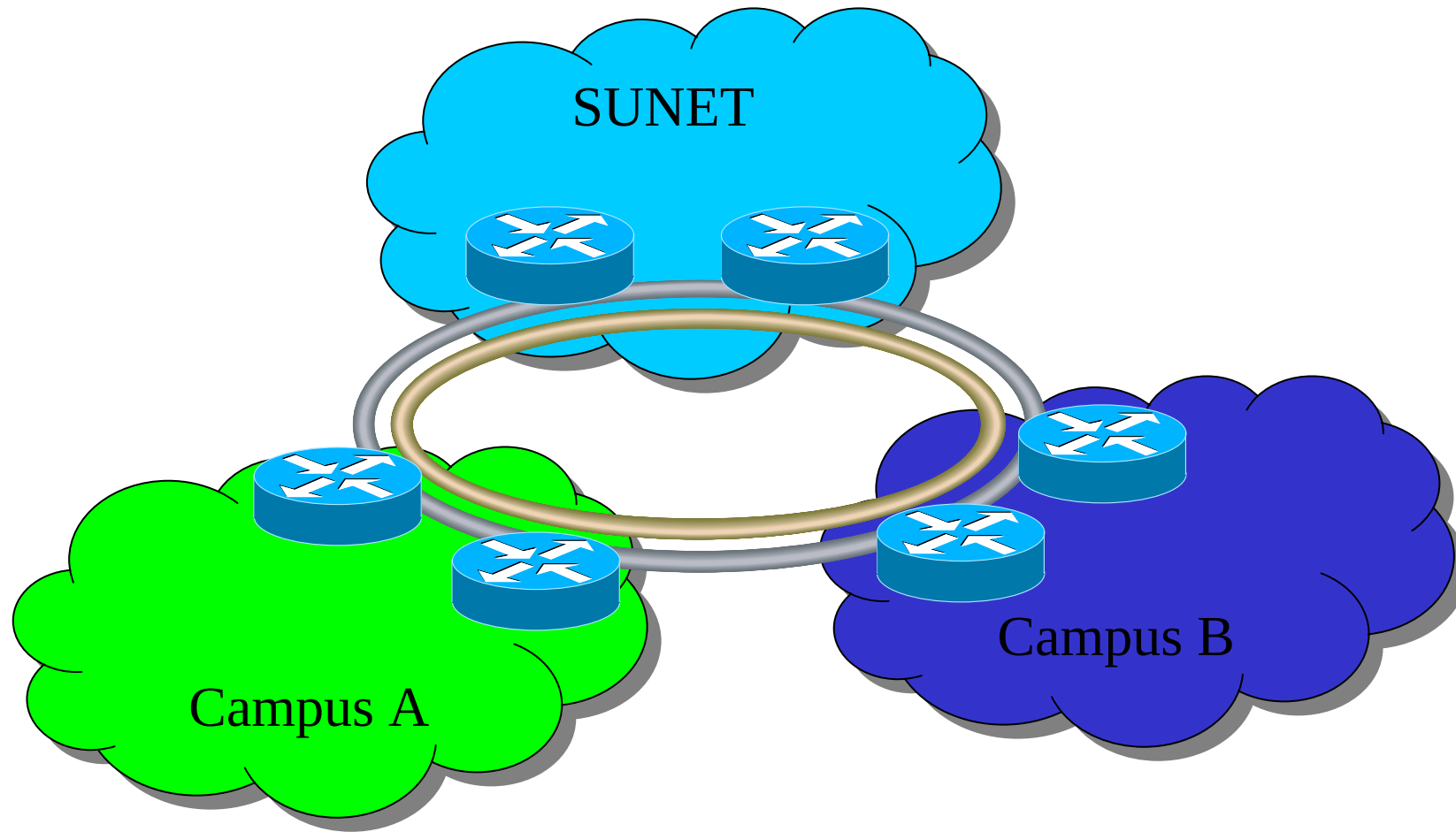
Lokalringen sitter på SRP2/0 (utom i Stockholm)

① Router 1 på orten

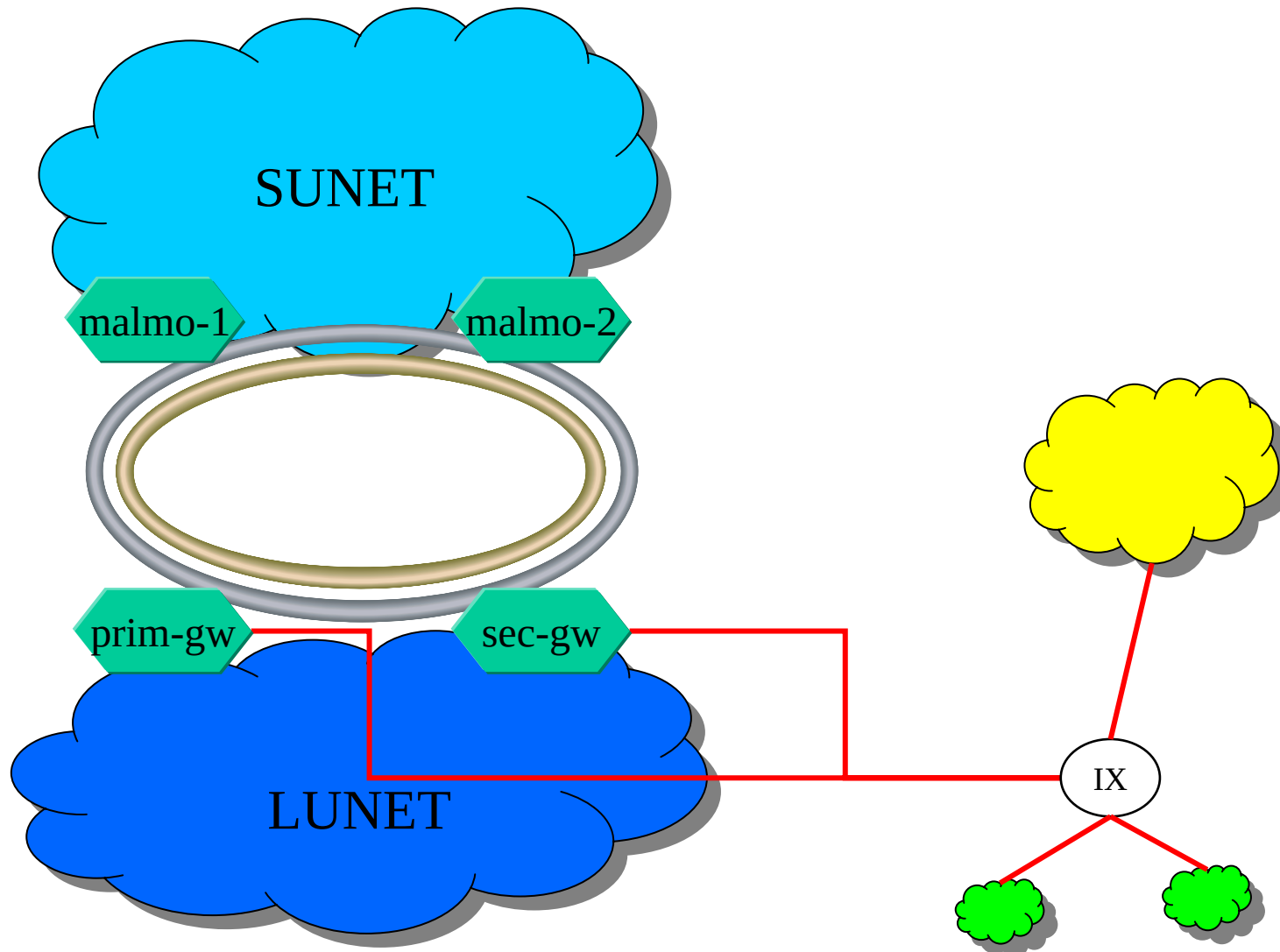
② Router 2 på orten



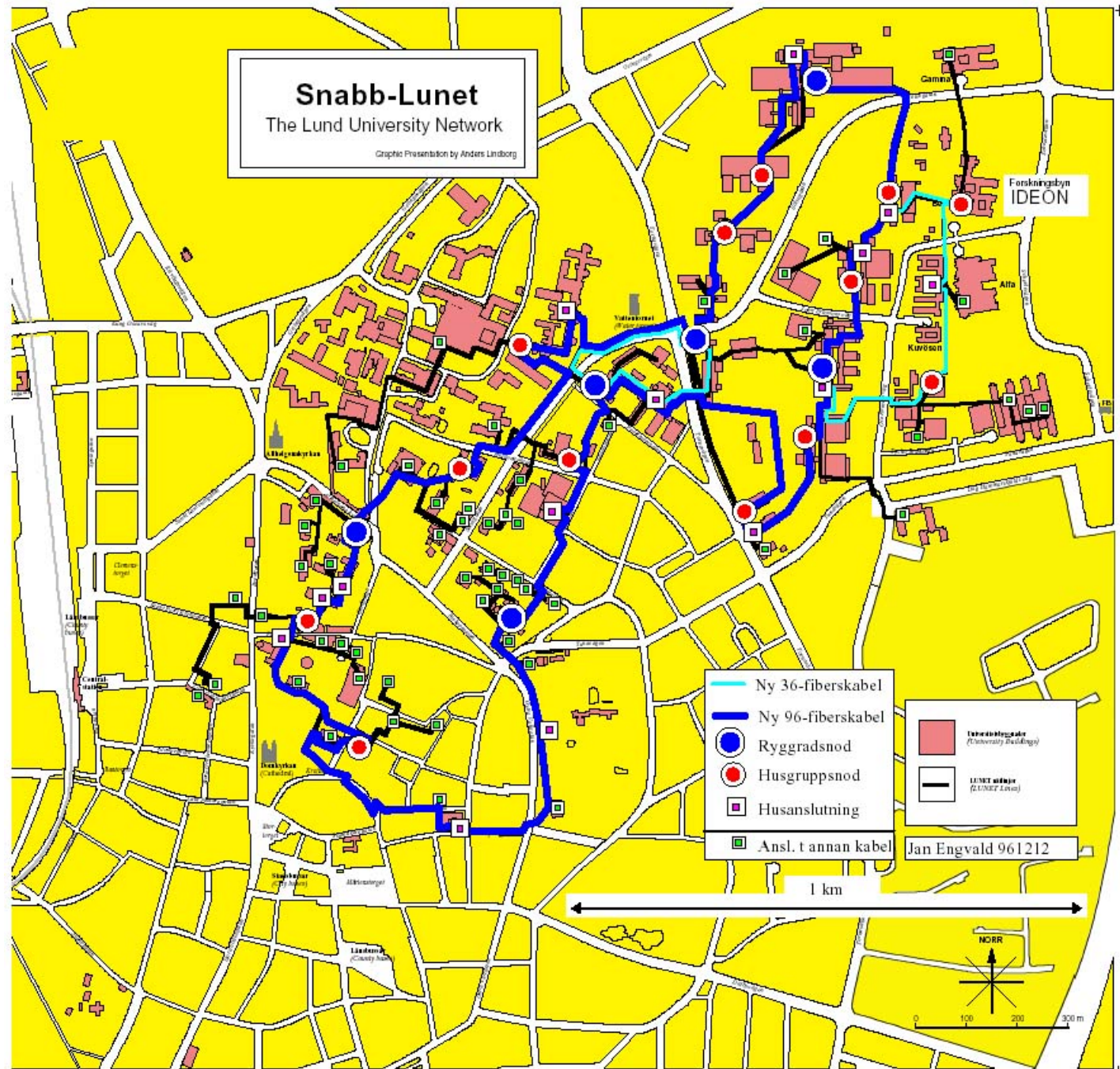
GigaSUNET, access network



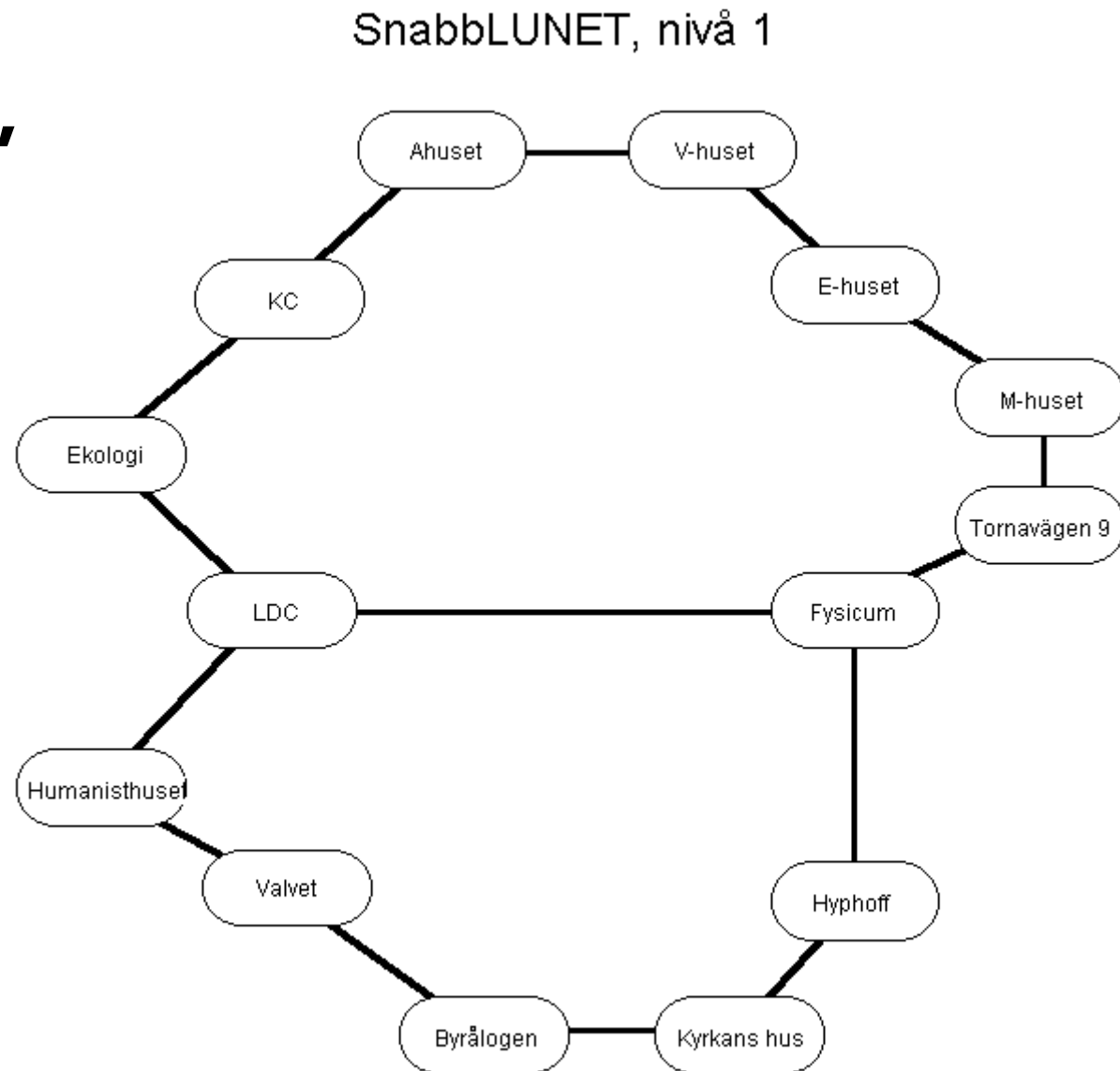
SnabbLUNET: inter network connections



SnabbLUNET: Backbone Optical Fibre in Lund

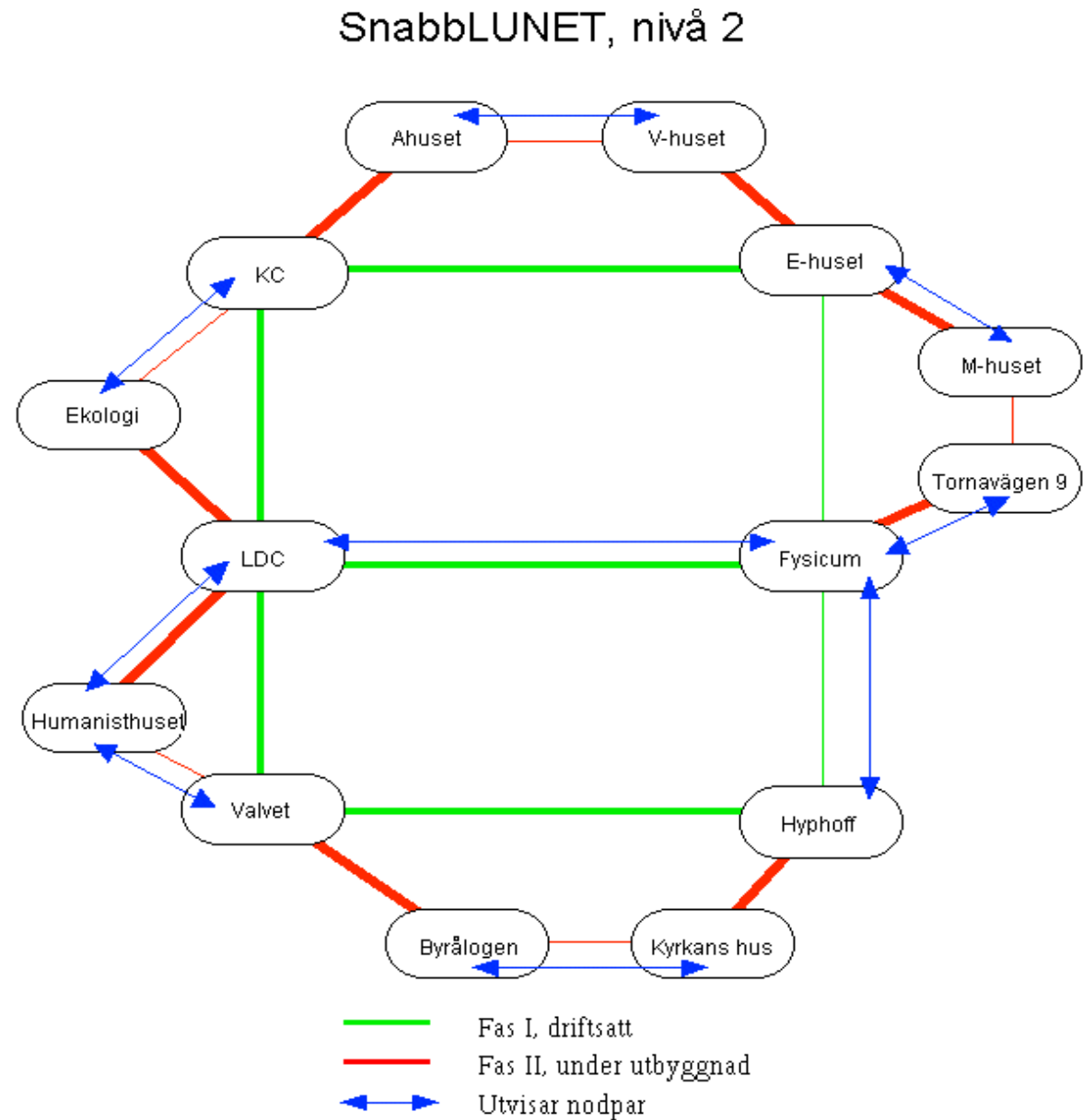


SnabbLUNET, Layer 1



Figur 1: Schematisk bild av fiberförläggningen i SnabbLUNET

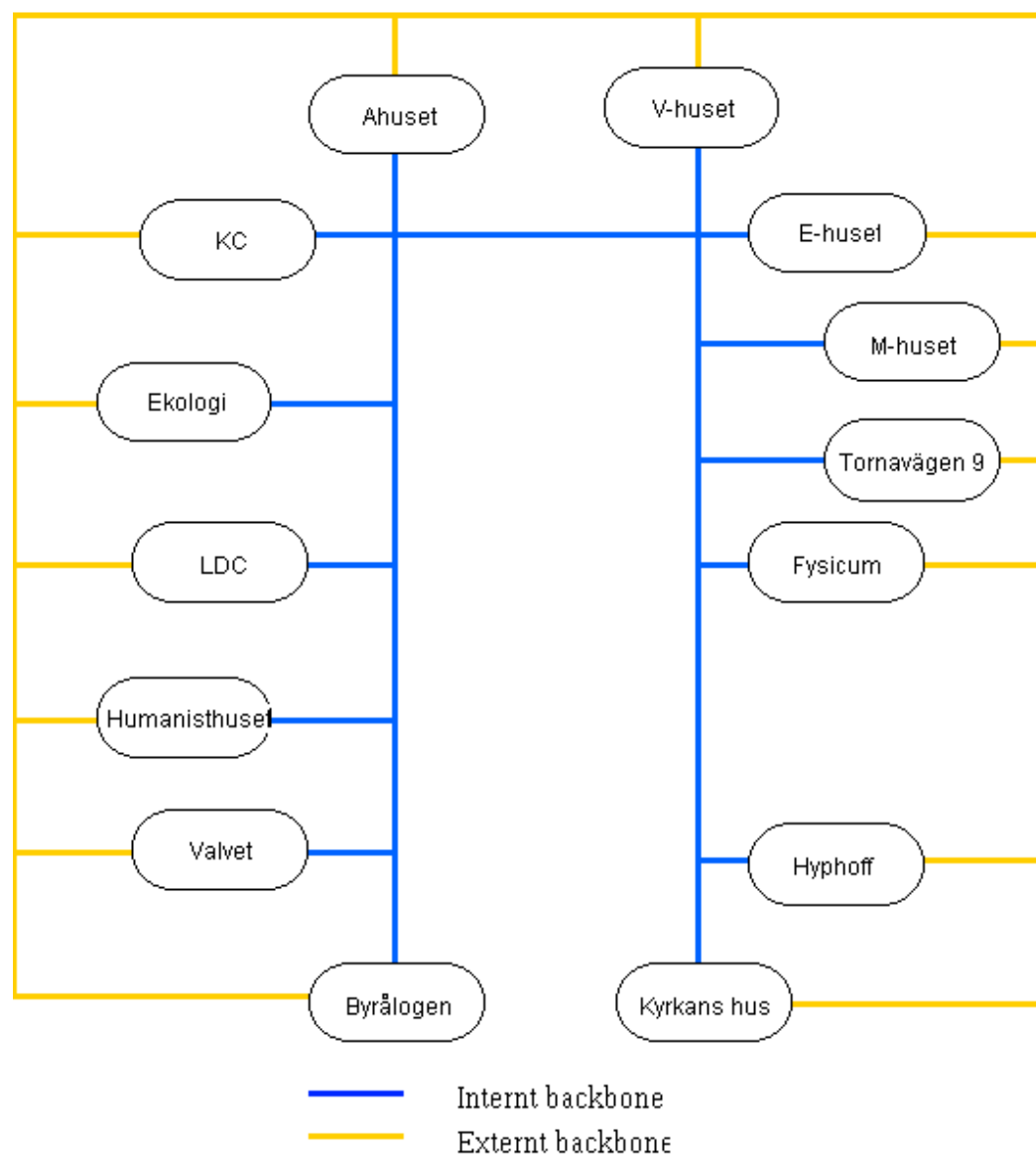
SnabbLUNET, Layer 2 (LU core only)



Figur 2: Schematisk bild av SnabbLUNETs nivå 2, och de nodpar som institutionsnäten ansluter till. Spanning Tree har normalt blockerat de förbindelser som är ritade med tunna streck.

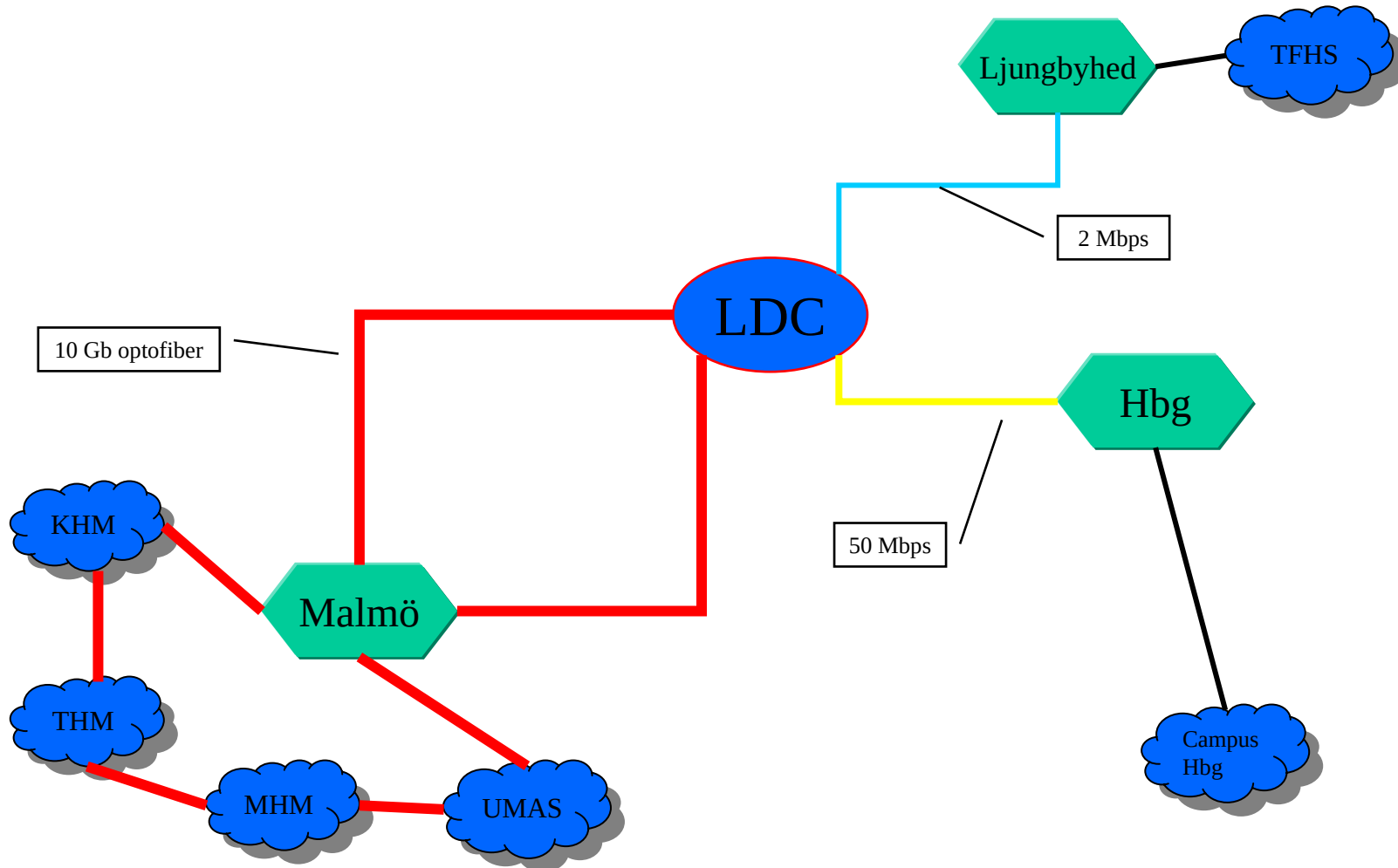
SnabbLUNET, Layer 3 (LU core only)

SnabbLUNET, nivå 3

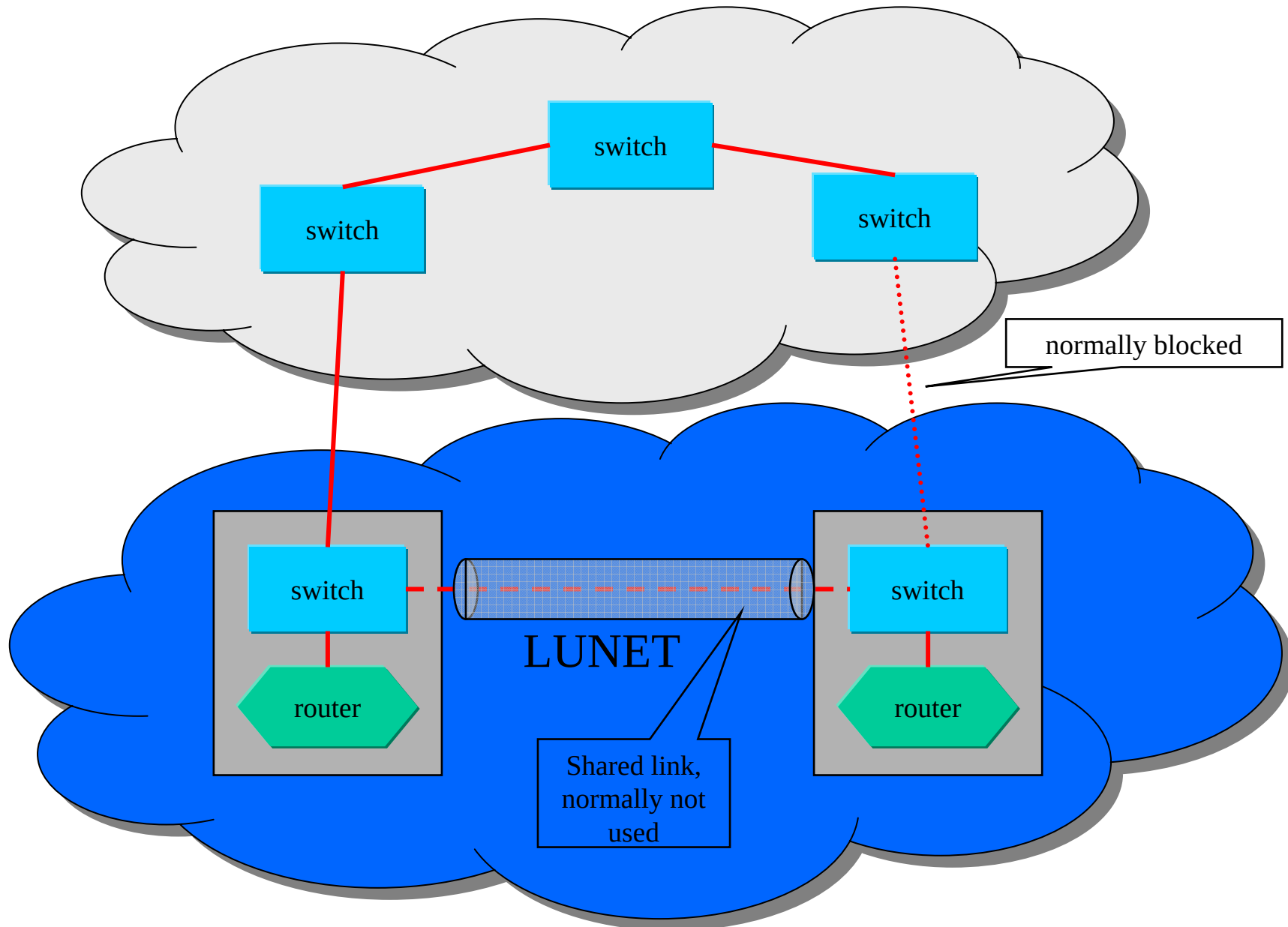


Figur 3: Schematisk bild av SnabbLUNETs nivå 3. Alla noder är anslutna till de båda backbone.

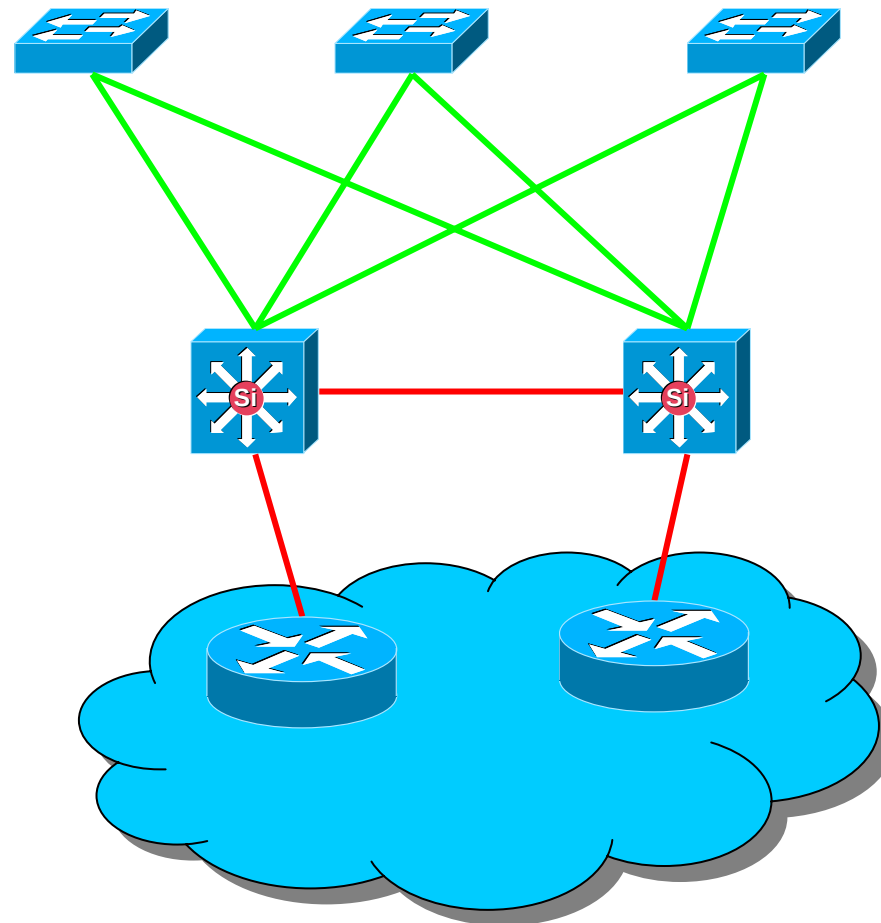
LUNET: WAN connections



Access Network



Access Network: Cisco's model



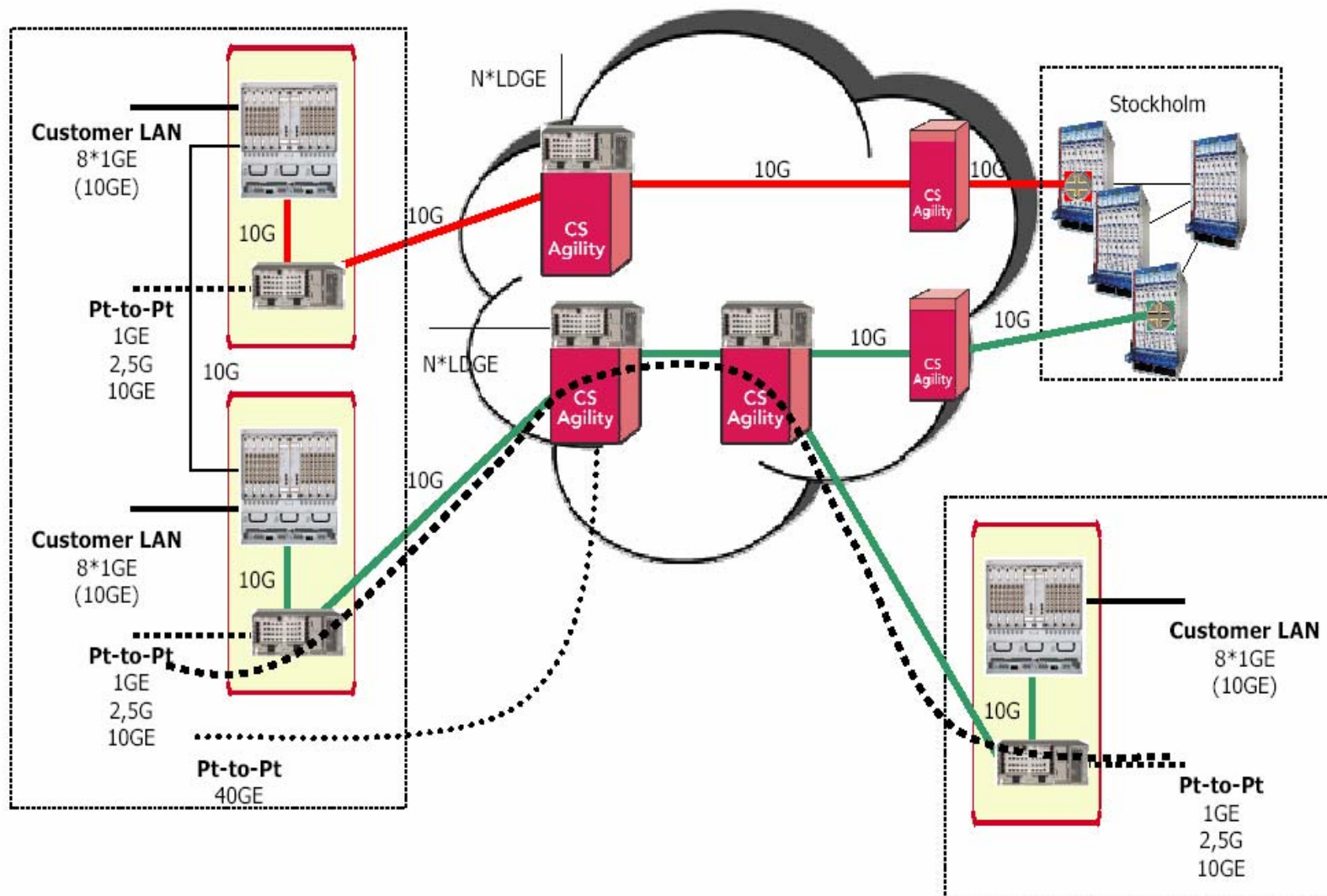
New technologies

- Ethernet with very high capacity (1-10 Gbps)
- Cheaper network adapters for long distance communication
- Wavelength multiplexing on optic fibres
 - “wavelengths/colours” , WDM
- L3-L4 switches
- Centralized routing – Collapsed backbone
 - Back to the future ...

Or



OptoSUNET



OptoSUNET

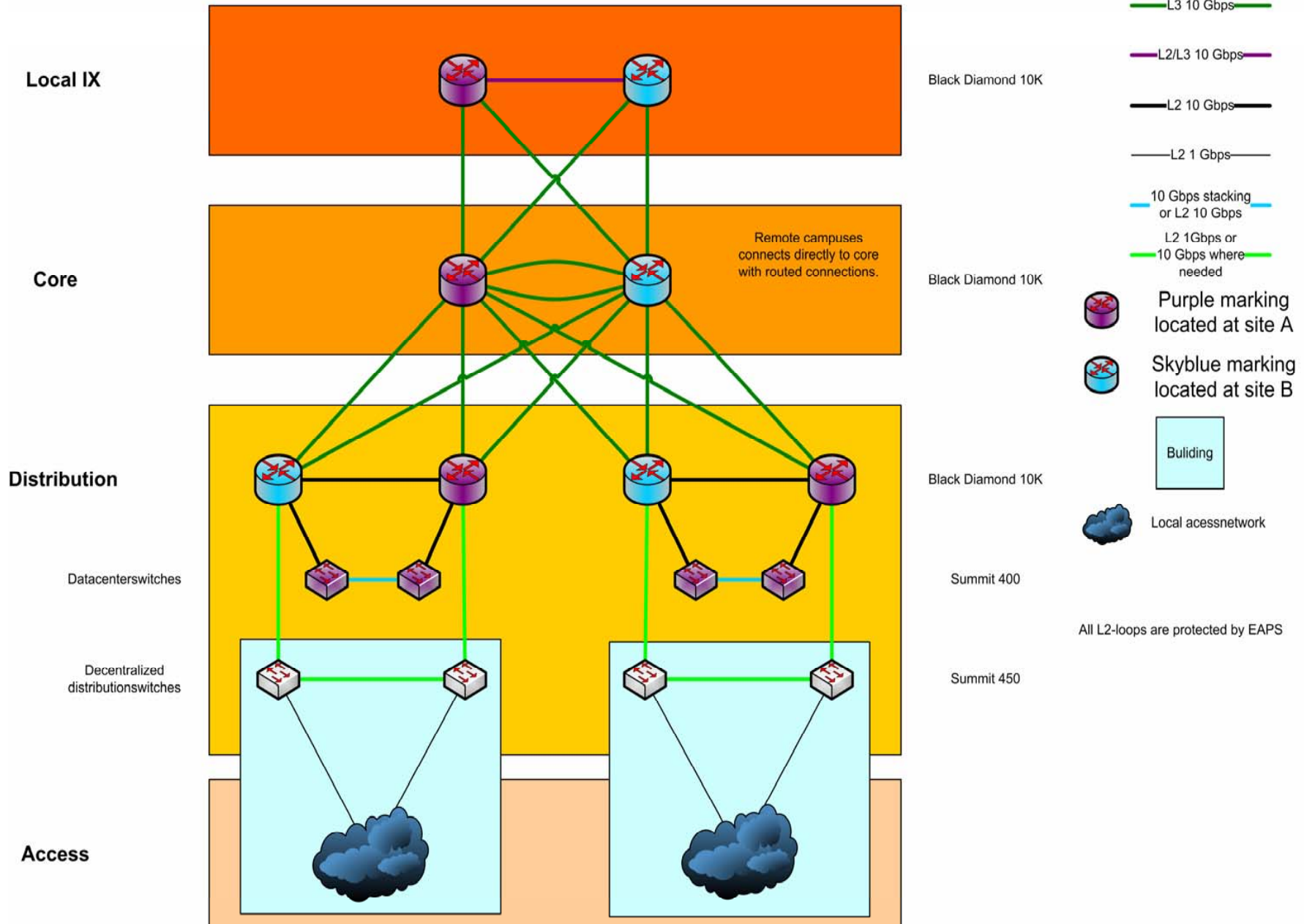
- 2 * 1, 10 Gbps to universities
- Hybrid network
 - Native IP/Internet
 - Wavelengths/DWDM
 - < 40 Gbps/colour)
- Fibre
 - TDC Song
- Active equipment
 - Owned by SUNET
 - Brand/Type to be decided

GigaLUNET

- 10 Gbps in Core and Centralized Distribution
- 1 – 10 Gbps in Decentralized Distribution and Access
- Single Mode Fibre
- EAPS in L2 (Distribution)
- Equal Path Cost in L3 (Core)

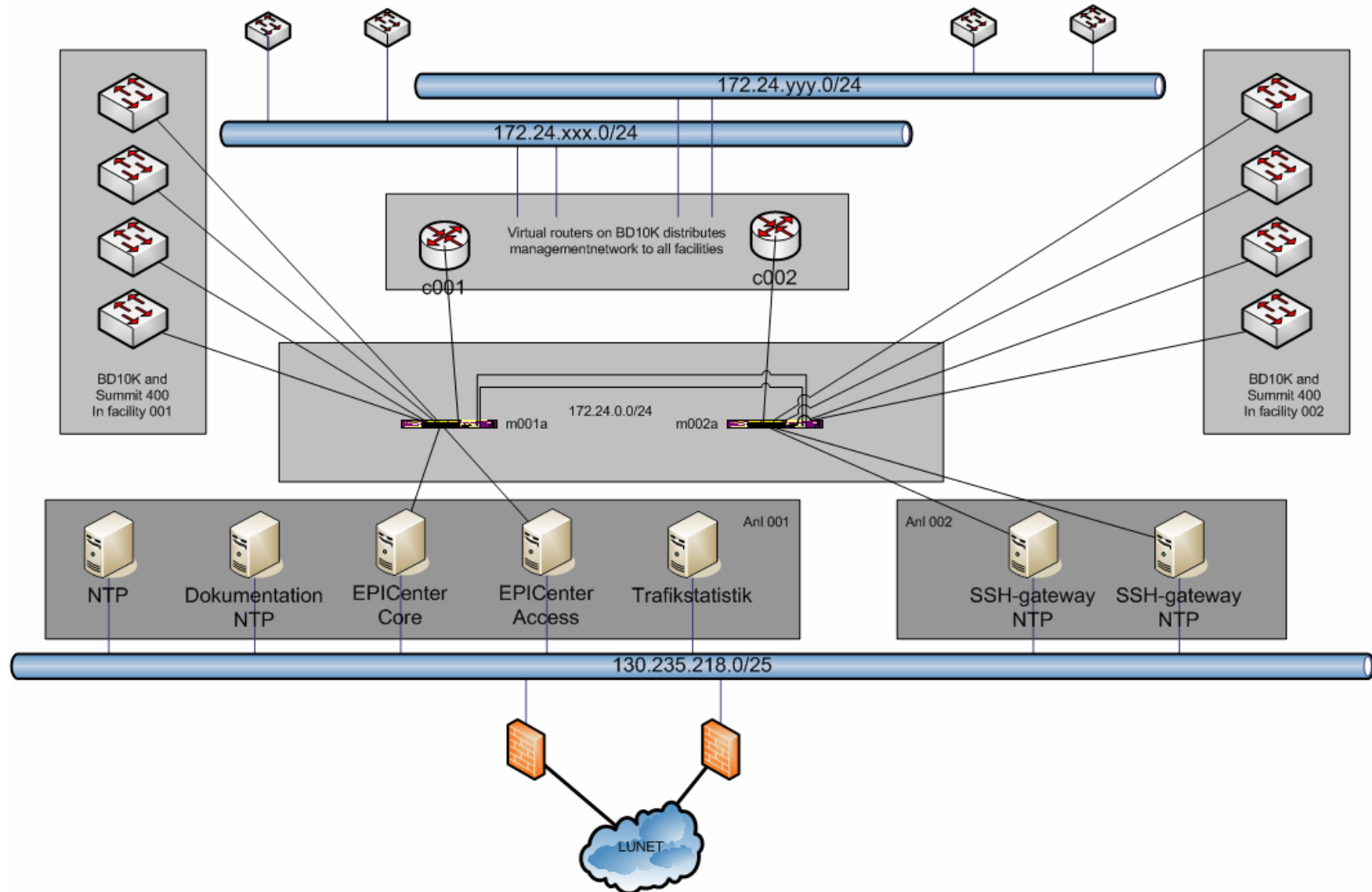
GigaLUNET design Campus Lund

2005-08-19 06:30 / mjh

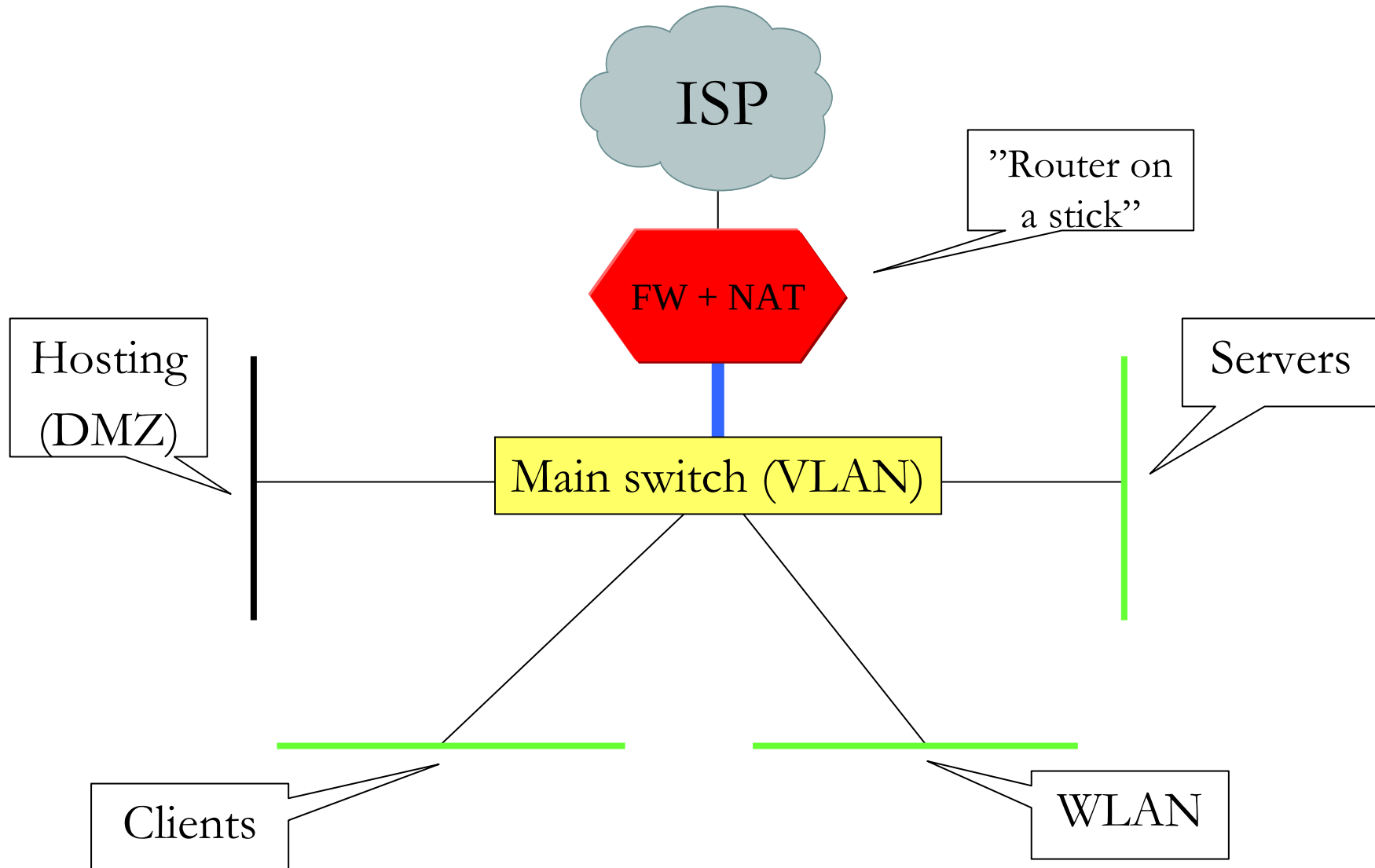


GigaLUNET design Managementnät

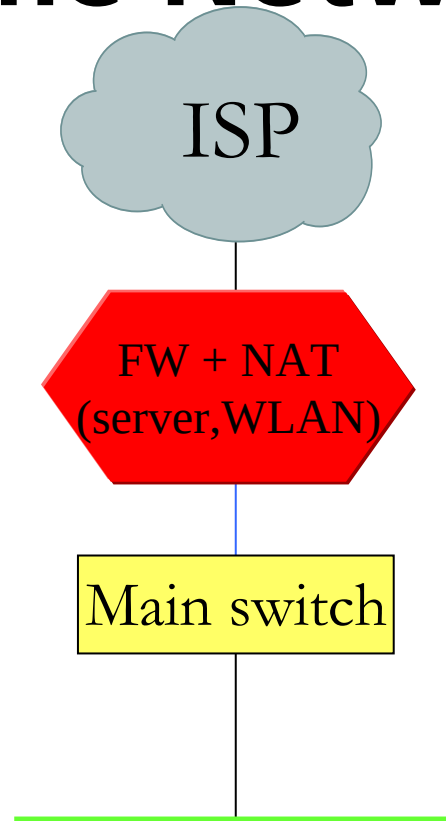
2006-01-13 08:30 / mjh



Smaller Enterprise Network



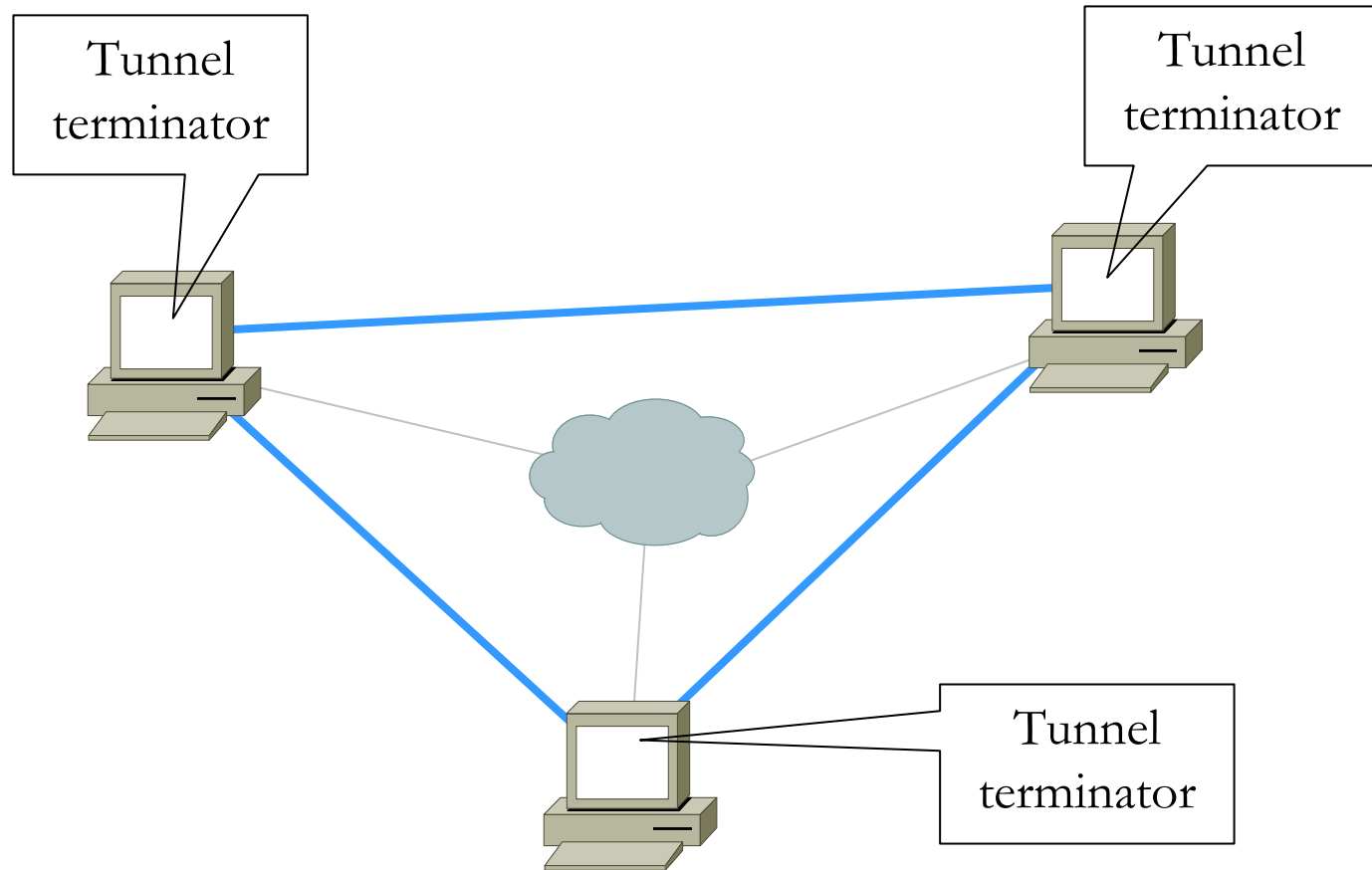
Home Network



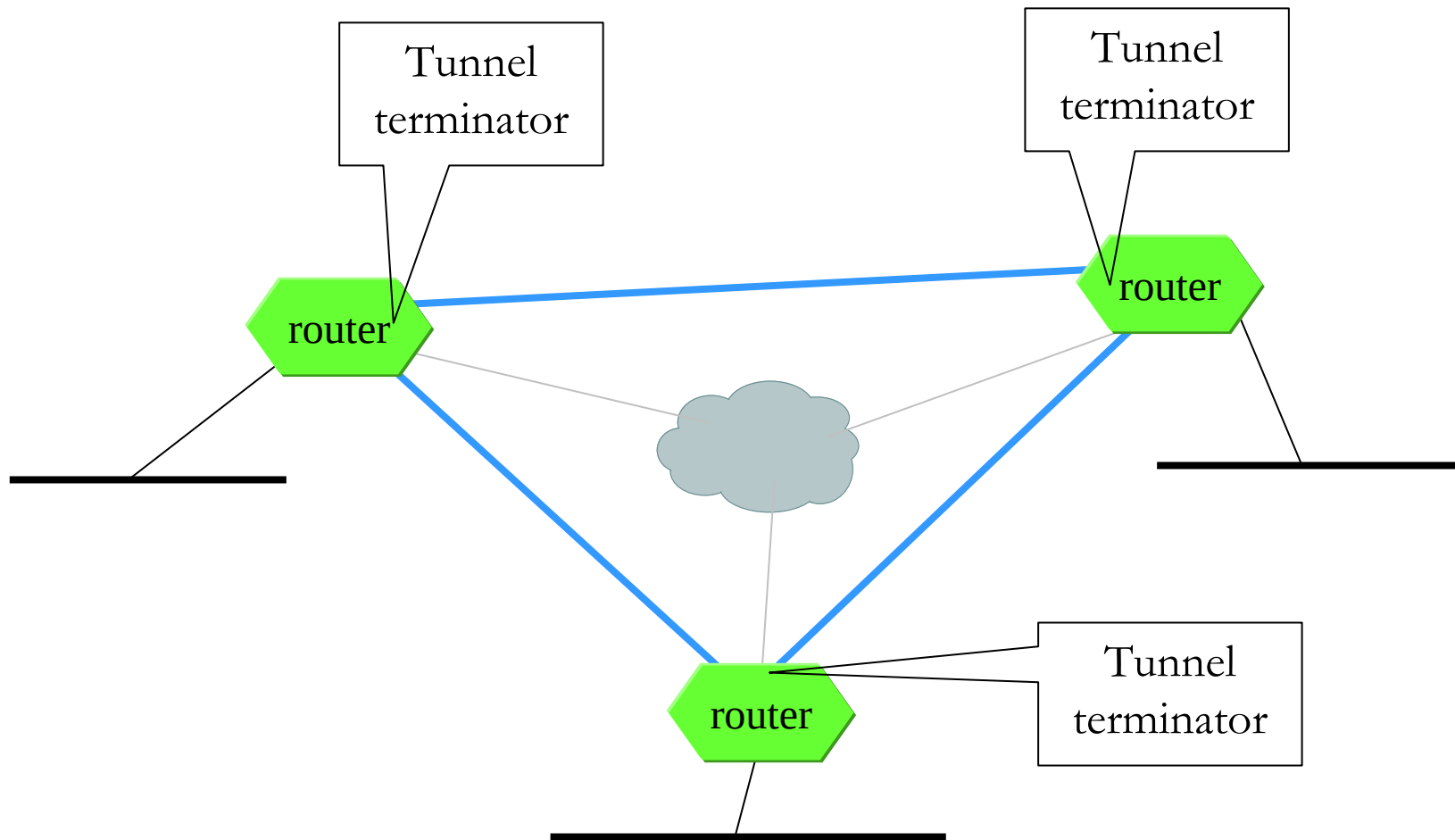
The Virtual Network

- Logical overlay network
- Built with VPN, MPLS tunnels
 - Host-to-Host
 - Network-to-Network
 - Combination

Host-to-Host VPN



Network-to-Network VPN



History: Links

- <http://www.zakon.org/robert/internet/timeline/>
- <http://www.isoc.org/internet/history/brief.shtml>
- http://basun.sUNET.se/html_docs/info_sUNET/historia.html
- <http://www1.ldc.lu.se/div/LDC-Historia.html>